

Integration over a Vanishing Cycle

P. Deligne

Introduction

Let k be a commutative field and let f be a formal power series in two indeterminates x and y over k :

$$f = \sum a_{n,m} x^n y^m.$$

Let $I(f)$ be the formal power series in one indeterminate t

$$I(f) = \sum a_{n,n} t^n.$$

The aim is to prove the following result.

Theorem 1. *If f represents an algebraic function of x and y , and if k has characteristic $p \neq 0$, then $I(f)$ represents an algebraic function of t .*

For a formal series in N indeterminates

$$g = \sum a_n x^n,$$

one defines similarly

$$I_N(g) = \sum a_{n,\dots,n} t^n,$$

and one has the following consequence.

Corollary 1. *If g is algebraic over the field $k(x_1, \dots, x_N)$, and if k has characteristic $p > 0$, then $I_N(g)$ is algebraic over $k(t)$.*

Under the stronger hypothesis $g \in k(x_1, \dots, x_N)$ this result is due to H. Furstenberg. The first purpose here is to give a geometric explanation of Furstenberg's result.

Suppose first that $k = \mathbb{C}$ and that the formal series f is convergent. Then $I(f)$ is convergent and has the integral representation

$$I(f)(t) = \frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt}. \quad (1)$$

Here $\pi : \mathbb{C}^2 \rightarrow \mathbb{C}$ is $(x, y) \mapsto xy$. The expression $(dx \wedge dy)/dt$ is a relative holomorphic differential on $\mathbb{C}^2 - \{(0, 0)\}$, relative to π ; it is also a section of the relative dualizing sheaf at $(0, 0)$, in the sense of Grothendieck duality. In the coordinate y on the fibre $xy = t$ one has

$$\frac{dx \wedge dy}{dt} = \frac{d(t/y) \wedge dy}{dt} = \frac{dy}{y},$$

up to the sign fixed by the chosen order of the variables.

If f converges in a polydisc of radius R , and if $|t| < R^2$, the cycle of integration is $|y| = r$, $|x| = |t|/r$, $xy = t$, with $|t|/R < r < R$. The integral is independent of r . Thus

$$\frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt} = \frac{1}{2\pi i} \int f(t/y, y) \frac{dy}{y},$$

and expansion into the series followed by the residue formula leaves only the terms with equal exponents in x and y .

The integration cycle is the Lefschetz vanishing cycle for π near $(0, 0)$: if $r = |t|^{1/2}$ the cycle visibly collapses as $|t| \rightarrow 0$. This explains the title.

Formula (1) suggests that the operation I has an intrinsic meaning for a morphism of formal schemes $\pi : X \rightarrow S$ over k with $X \simeq \mathrm{Spf} k[[x, y]]$, $S \simeq \mathrm{Spf} k[[t]]$, and π having a non-degenerate quadratic singularity. The two branches of $\pi^{-1}(0)$ are assumed to be defined over k , and an order between them is chosen. Thus we choose coordinates with $\pi(x, y) = xy$, with the first branch of $xy = 0$ being $x = 0$.

Let $\Omega_{X/k}^1$ be the Kähler differentials defined with respect to the adic topology. In coordinates $\Omega_{X/k}^1$ is freely generated by dx, dy , and $\Omega_{S/k}^1$ by dt . Let

$$\omega_{X/k} = \det \Omega_{X/k}^1, \quad \omega_{S/k} = \Omega_{S/k}^1, \quad \omega_{X/S} = \omega_{X/k} \otimes \pi^* \omega_{S/k}^{-1}.$$

In coordinates $\omega_{X/S}$ is generated by $(dx \wedge dy)/dt$. The operation I will be interpreted as a morphism

$$\pi_* \omega_{X/S} \longrightarrow \mathcal{O}_S,$$

or, for $S = \mathrm{Spf}(V)$, as a V -linear map

$$H^0(X, \omega_{X/S}) \longrightarrow V.$$

Section 1 provides this interpretation algebraically. The integration sign in (1) is replaced by Grothendieck's local trace. For a uniformizer t of S , the reduction modulo t^n is denoted with the index n . Let A_n be the complete local ring whose formal spectrum is the reduction X_n ; in coordinates $A_n = k[[x, y]]/(xy)^n$. It is convenient to replace X_n by the ordinary scheme $X'_n = \mathrm{Spec} A_n$. Let O be its closed point. Grothendieck's local trace is a morphism

$$\mathrm{Tr} : H_{\{O\}}^1(X'_n, \omega) \longrightarrow H^0(S_n, \mathcal{O}).$$

The role of the vanishing cycle is played by an element

$$v_n \in H_{\{O\}}^1(X'_n, \mathcal{O}).$$

In coordinates v_n comes from $\mathrm{Ext}_{A_n}^1(A_n/(x+y)^{2n-1}, A_n)$. In Section 2 these classes are related to an element of the Lie algebra of a Picard scheme. Section 3 globalizes the construction and proves the theorem and the corollary.

1 Algebraic analogue of the integral formula

Keep the notation above. The inclusion $X'_1 \subset X'_n$ is a homeomorphism, since X'_1 is defined by a nilpotent ideal in X'_n . The complement of the origin in X'_n has two connected components corresponding to the branches $x = 0$ and $y = 0$ of X'_1 . Let u_n be the section of the structure sheaf of $X'_n - \{O\}$ which is 1 on the first branch and 0 on the second.

Algebraically, the homomorphism

$$A_n = k[[x, y]]/((xy)^n) \rightarrow k[[x, y]]/(x^n) \times k[[x, y]]/(y^n)$$

induces an isomorphism after inverting $x + y$, and u_n is the element whose image is $(1, 0)$. Let v_n be the image of u_n under the boundary map

$$H^0(X'_n - \{O\}, \mathcal{O}) \longrightarrow H_{\{O\}}^1(X'_n, \mathcal{O}).$$

Applying $\mathrm{Ext}^\bullet(-, A_n)$ to

$$0 \rightarrow ((x + y)^N) \rightarrow A_n \rightarrow A_n/((x + y)^N) \rightarrow 0$$

gives a diagram showing that, for $N = 2n - 1$, v_n lies in the image of

$$\mathrm{Ext}_{A_n}^1(A_n/((x + y)^{2n-1}), A_n).$$

1.2. Proposition. For $f \in k[[x, y]]$ and for every n ,

$$\mathrm{Tr} \left(f \frac{dx \wedge dy}{dt}, v_n \right) = -I(f) \pmod{t^n}. \quad (1.2.1)$$

The trace is Grothendieck's local trace. We prove the formula when k is infinite; the general case follows by extension of scalars. Let $I'(f)$ denote the left-hand side of (1.2.1). It is a continuous $k[[t]]/(t^n)$ -linear map $A_n \rightarrow k[[t]]/(t^n)$. For $\lambda, \mu \in k^*$ let $\sigma(\lambda, \mu)$ be the automorphism

$$(x, y, t) \mapsto (\lambda x, \mu y, \lambda \mu t).$$

It preserves $(dx \wedge dy)/dt$ and v_n , hence

$$I'(\sigma(\lambda, \mu) f) = \sigma(\lambda, \mu)^* I'(f). \quad (1.2.2)$$

Applying this to monomials gives $I'(x^a y^b) = 0$ unless $a = b$. By continuity,

$$I'(f) = I(f)I'(1).$$

The scalar $I'(1)$ is independent of n , so it remains to compute it for $n = 1$.

For $n = 1$, $X'_1 = \mathrm{Spec} k[[x, y]]/(xy)$ is the union of two smooth formal arcs meeting transversely. Sections of ω are pairs of meromorphic differentials (α, β) on the two branches with at most simple poles at O and residues summing to zero. On the punctured curve arbitrary pole orders are allowed. The composite

$$H^0(X'_1 - \{O\}, \omega) \rightarrow H^1_{\{O\}}(X'_1, \omega) \xrightarrow{\mathrm{Tr}} k$$

is the negative of $(\alpha, \beta) \mapsto \mathrm{Res} \alpha + \mathrm{Res} \beta$. In these terms $(dx \wedge dy)/dt$ and u_1 give residue sum 1, proving (1.2.1).

2 Relation with the Picard functor

Let $S = \mathrm{Spec} k[[t]]$, let s be its closed point and η its generic point. Let $\pi : Y \rightarrow S$ be proper and flat, purely of relative dimension one, and let $O \in Y_s(k)$. Assume that the completion of Y at O is the formal singularity considered above. Thus Y is regular at O , O is an isolated non-smooth point of π , the singularity is non-degenerate quadratic, and the two branches are defined over k and ordered.

We may further suppose Y regular and connected. Let Y_η be the generic fibre over $k((t))$.

2.2. Lemma. The curve Y_η is absolutely irreducible and generically smooth over $k((t))$.

Indeed, the special fibre is connected and has a rational point, hence is absolutely connected. It is smooth near O away from O , so after a separable extension it has smooth rational points. The corresponding base change has a section; regularity then implies irreducibility, and absolute connectedness gives absolute irreducibility. The smooth locus of the generic fibre is non-empty and therefore dense.

Let C be a complete reduced generically smooth curve over a field K , and D a Cartier divisor on C . The generalized Jacobian $\mathrm{Pic}(C; D)$ represents invertible sheaves on C_T trivialized on D_T . It is smooth, and

$$\mathrm{Lie} \mathrm{Pic}(C; D) = H^1(C, \mathcal{O}(-D)). \quad (2.3.1)$$

If C is smooth, the map

$$j : C - D \rightarrow \text{Pic}(C; D), \quad u \mapsto [\mathcal{O}(u(T))] \quad (2.3.2)$$

relates Serre duality with the pullback of invariant differentials on the Picard scheme.

Returning to Y/S , let Z be a relative Cartier divisor disjoint from O , and let $\text{Pic}(Y; Z)^0$ be the relative generalized Picard scheme parametrizing line bundles on Y trivialized along Z and of degree zero on each geometrically irreducible component of each fibre. For each n one constructs morphisms

$$H^0(\mathcal{O}_T, \mathcal{O}_T) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(-Z_{n,T})), \quad (2.4.6)$$

$$H^0(\mathcal{O}_T, \mathcal{O}_T^*) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(1 \text{ on } Z_{n,T})^*). \quad (2.4.7)$$

The multiplicative version gives a morphism of $k[[t]]/(t^n)$ -group schemes

$$w_n : \mathbf{G}_m \longrightarrow \text{Pic}(Y_n, Z)^0. \quad (2.4.8)$$

For $m < n$, the reduction of w_n modulo t^m is w_m ; for $n = 1$ this is the morphism associated with the two branches.

Although one cannot take a direct limit to get a morphism $\mathbf{G}_m \rightarrow \text{Pic}(Y; Z)^0$, one can restrict to finite subgroup schemes. For every integer m one obtains

$$w : \mu_m \longrightarrow \text{Pic}(Y, Z)^0. \quad (2.4.9)$$

Passing to Lie algebras gives

$$dw : \text{Lie } \mathbf{G}_m \longrightarrow \text{Lie } \text{Pic}(Y; Z)^0, \quad (2.4.10)$$

and hence an element

$$dw(1) \in H^1(Y, \mathcal{O}(-Z)). \quad (2.4.11)$$

Let $\omega_{Y/S}$ be the relative dualizing sheaf and let φ be a section of $\omega_{Y/S}(Z)$. In formal coordinates near O ,

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

The restriction of φ to the generic fibre is paired with $H^1(Y_\eta, \mathcal{O}(-Z_\eta))$ by Serre duality.

2.5. Proposition. With the notation above,

$$I(f) = -(dw(1), \varphi). \quad (2.5.1)$$

The right-hand side lies in $k[[t]]$, not merely in $k((t))$. Reducing modulo t^n , the local trace and the global Serre trace are compatible; the proposition is reduced to the following lemma.

2.6. Lemma. The morphism from local cohomology at O to $H^1(Y_n, \mathcal{O}(-Z_n))$ sends v_n to $dw_n(1)$.

The image of v_n is also the image of 1 under (2.4.6). The assertion follows from the description of dw_n via dual numbers over $k[[t]]/(t^n)$ and the natural exact sequence comparing the additive and multiplicative constructions.

3 Proof of the theorem

Let S_0 be a smooth curve over k , let $s \in S_0(k)$, and let S be the completion at s . Let $\pi : Y_0 \rightarrow S_0$ be proper and flat, purely of relative dimension one, with Y its base change to S and with a point $O \in Y_s$ such that $(Y/S, O)$ is as above. We suppose, for simplicity, that Y_0 is regular and that π has geometrically connected fibres. Let ω_{Y_0/S_0} be the relative dualizing sheaf.

Let φ be a rational section of ω_{Y_0/S_0} , regular at O . In formal coordinates it can be written

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

3.2. Proposition. If k has characteristic $p > 0$, then $I(f)$ is algebraic over the field $k(S_0)$ of rational functions on S_0 .

After multiplying φ by a rational function on S_0 vanishing sufficiently at s , and after shrinking S_0 , we may assume that for a suitable relative Cartier divisor Z_0 disjoint from O , the form φ is a section of ω on $Y_0 - Z_0$. Let $\text{Pic}(Y_0; Z_0)^0$ be the relative generalized Picard scheme. Via Serre duality, φ corresponds to a section of the dual of the vector bundle

$$\text{Lie Pic}(Y_0; Z_0)^0 = R^1 \pi_* \mathcal{O}(-Z_0).$$

After base change to S we have $dw : \mathcal{O} \rightarrow \text{Lie Pic}(Y; Z)^0$, and Proposition 2.5 reduces Proposition 3.2 to the following key lemma.

3.3. Key Lemma. If $p > 0$, then dw is algebraic over S_0 .

In characteristic p , the inclusion $\mu_p \subset \mathbf{G}_m$ induces an isomorphism on Lie algebras:

$$\text{Lie}(\mu_p) \xrightarrow{\sim} \text{Lie}(\mathbf{G}_m).$$

Modulo t^n , one checks that dw is the differential of the morphism

$$w : \mu_p \rightarrow \text{Pic}(Y; Z)^0.$$

It remains to use the rigidity of μ_p : for every flat commutative group scheme P over S_0 , the functor $\text{Hom}(\mu_p, P)$ is represented by an etale scheme over S_0 . Applying this to $P = \text{Pic}(Y_0; Z_0)^0$, the morphism w is pulled back from a universal morphism over an etale cover of S_0 . Hence its differential, and therefore dw , is algebraic.

Let $k\{x_1, \dots, x_N\}$ be the henselization of the local ring of affine N -space at the origin. Its completion is $k[[x_1, \dots, x_N]]$, and an element of the completion is algebraic over $k(x_1, \dots, x_N)$ if and only if it lies in the henselization.

Proof of the theorem

If f is algebraic, there exists a surface Y_0 with a rational point O , an etale morphism $h : Y_0 \rightarrow \mathbf{A}^2$ sending O to the origin, and a section whose formal expansion at O is f . Let $S_0 = \mathbf{A}^1$ with coordinate t , and let π' be the composite

$$Y_0 \xrightarrow{h} \mathbf{A}^2 \xrightarrow{xy} \mathbf{A}^1 = S_0.$$

The differential form $f h^*((dx \wedge dy)/dt)$ is a rational section of the relative dualizing sheaf. After compactifying and resolving singularities, one obtains a proper flat regular model to which Proposition 3.2 applies. Therefore $I(f)$ is algebraic.

Proof of the corollary

Proceed by induction on N . The cases $N = 0$ and $N = 1$ are immediate, and $N = 2$ is the theorem. For $N > 2$, write

$$g = \sum g_{n,m} x_{N-1}^n x_N^m, \quad g_{n,m} \in k[[x_1, \dots, x_{N-2}]],$$

and put

$$I(g) = \sum g_{n,n} x_{N-1}^n.$$

Then

$$I_N(g) = I_{N-1}(I(g)). \quad (3.6.1)$$

If g is algebraic, the coefficients $g_{n,m}$ are algebraic over $k(x_1, \dots, x_{N-2})$. Applying the theorem over the fraction field of the henselization in the first $N - 2$ variables shows that $I(g)$ is algebraic, and induction proves the claim.

Degree estimates and remarks

The proof gives an estimate. With the notation of the geometric construction, let α be the degree of the curve S_0 over the t -line, let g be the genus of the normalization of the geometric generic fibre, and let $z = \max(0, z' - 1)$ where z' is the number of geometric points of the divisor Z . Then the degree d of the equation satisfied by $I(f)$ is bounded by

$$d < \alpha p^{g+z}. \quad (3.7.1)$$

For a formal algebraic series with integer coefficients, this implies that for almost all primes p , the degree d_p of the reduction of $I(f)$ modulo p satisfies

$$d_p < \alpha p^{g+z}. \quad (3.7.2)$$

Using μ_{p^m} when $p^m = 0$ on the base gives analogous information modulo powers of p , for almost all primes.

A direct proof for N variables would be preferable. Analytically, for a convergent series g one expects the formula

$$I(g)(t) = \int_{Z(t)} g \frac{dz_1 \cdots dz_N}{dt}, \quad (1')$$

where

$$Z(t) = \{(x_1, \dots, x_N) : x_1 \cdots x_N = t, |x_i| = r_i, \prod r_i = |t|\}.$$

A direct proof of this kind would be a test case for a p -adic theory of vanishing cycles.

Furstenberg's proof in the rational case shows that if g is one component of a column vector h with entries in $k[[x_1, \dots, x_N]]$ satisfying

$$h = Ah^p$$

for a square matrix A with polynomial entries, then $I(g)$ has the same property and is algebraic. This type of equation recalls F -crystals in crystalline cohomology. Since $T_p \text{ Pic}$ is an avatar of p -adic H^1 , the proof here and Furstenberg's proof are perhaps less distant from each other than they first appear.

A Signs

The treatment of signs in the duality formalism for coherent algebraic sheaves is delicate. We record the conventions used here. For a flat relative complete intersection morphism $f : X \rightarrow S$, purely of relative dimension n , the relative dualizing sheaf $\omega_{X/S}$ is related to the relative dualizing complex by

$$K_{X/S} = \omega_{X/S}[n]. \quad (\text{S.1})$$

When X is smooth over S , $\omega_{X/S} = \Omega_{X/S}^n$. If X is the fibre product over S of morphisms $f_i : X_i \rightarrow S$ of relative dimensions n_i , then

$$K_{X/S} = \boxtimes_i K_{X_i/S}. \quad (\text{S.2})$$

Equivalently, after choosing a total order on the index set,

$$\omega_{X/S} = \bigotimes_i \omega_{X_i/S}, \quad (\text{S.3})$$

with the sign change $(-1)^N$ when the order is changed, where N is the sum of $n_i n_j$ over inverted pairs.

For $f : X \rightarrow S$ proper as above, the trace is

$$\text{Tr} : Rf_* K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.4})$$

It induces

$$\text{Tr} : R^n f_* \omega_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.5})$$

For a section s of f there is a local trace

$$\text{Tr} : Rs^! K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.6})$$

The choices are fixed by requiring compatibility between global and local traces and by normalizing, for a smooth curve X over an algebraically closed field and a point $x \in X(k)$, the local trace

$$H_{\{x\}}^1(X, \Omega^1[1]) \rightarrow k \quad (\text{S.7})$$

to be the residue map Res_x .

With these conventions, for a smooth complete curve X and $x \in X(k)$, the boundary map

$$H^0(X - \{x\}, \Omega^1) \rightarrow H_{\{x\}}^1(X, \Omega^1)$$

composed with the trace is $-\text{Res}_x$. For products, if $X = X_1 \times X_2$ and $u_i \in H^{n_i}(X_i, \omega_{X_i})$, then

$$\text{Tr}(\text{pr}_1^* u_1 \smile \text{pr}_2^* u_2) = (-1)^{n_1 n_2} \text{Tr}(u_1) \text{Tr}(u_2).$$

Over $k = \mathbf{C}$, comparison with the transcendental trace identifies the algebraic trace with integration over $X(\mathbf{C})$, with the standard normalization by $(2\pi i)^{-n}$. For projective space $\mathbf{P}^n$ with homogeneous coordinates $T_0, \dots, T_n$ and $t_i = T_i/T_0$, the Čech class of

$$\frac{dt_1 \wedge \dots \wedge dt_n}{t_1 \dots t_n}$$

has trace $(-1)^{n(n+1)/2}$ with these conventions.

References

- [1] H. Furstenberg, Algebraic functions over finite fields, *J. Algebra* **7** (1967), 271–277.
- [2] M. Raynaud, Specialisation du foncteur de Picard, *Publ. Math. IHES* **38** (1970), 27–76.
- [3] M. Raynaud, *Anneaux locaux henseliens*, Lecture Notes in Mathematics 169, Springer, 1970.

The Lemma of Gabber

Pierre Deligne

The principal result of this exposé, Theorem 1.4, is due to O. Gabber.

1 Statements

1.1 In this exposé we systematically work with algebraic spaces, rather than only with schemes. This raises a problem of references. We shall allow ourselves to refer to an article or theorem proved for schemes and apply it to algebraic spaces whenever the same proof also proves the assertion for algebraic spaces. A proof becomes problematic when it uses the fact that every point has an affine neighbourhood and when “neighbourhood” cannot be replaced by “étale neighbourhood”.

This leads us to speak of an *abelian algebraic space* A over S , where one would ordinarily speak of an abelian scheme over S : this means a commutative S -algebraic space in groups

$$f : A \longrightarrow S$$

with f proper, flat, and of finite presentation, and with geometric fibres abelian varieties. We shall call A a *semi-abelian algebraic space* over S if it is a commutative S -algebraic space in groups $f : A \rightarrow S$, with f separated, flat, of finite presentation, and with geometric fibres extensions of abelian varieties by tori.

Note, however, the following points.

- (a) If S is the spectrum of a field, every algebraic space in groups over S is a scheme, and indeed is quasi-projective.
- (b) If S is a trait, every algebraic space in groups G over S , locally of finite type and separated over S , is a scheme. This is proved for the smooth case in Raynaud’s work on Néron models.
- (c) If S is normal integral, with generic point η , and if A is an abelian algebraic space over S , Grothendieck’s arguments show that every polarization of A_η extends uniquely to a polarization of A over S . It follows that A is projective over S , hence in particular that A is a scheme if S is a scheme.

An algebraic space $f : X \rightarrow S$, flat of finite presentation, separated over S , and with proper and connected geometric fibres, is proper over S . By localization on S and passage to limits, one reduces to the case where S is the spectrum of a noetherian local ring; by faithfully flat descent one may suppose the ring complete, and then one applies EGA III, first part, 5.5. In particular, a semi-abelian algebraic space over S all of whose fibres are abelian varieties is abelian over S .

Unless explicitly stated otherwise, all schemes and algebraic spaces considered below are quasi-compact and quasi-separated.

1.2 Let U be a dense open subspace of a normal algebraic space S , and let A be an abelian algebraic space over U . A *semi-abelian prolongation* of A over S is a semi-abelian algebraic space P over S together with an isomorphism

$$A \xrightarrow{\sim} P|_U.$$

Theorem 1.1 (Raynaud). *Under the preceding hypotheses, if A admits a semi-abelian prolongation, then that prolongation is unique up to unique isomorphism.*

Localization and passage to limits reduce the proof to the case where S is a noetherian scheme. If S is a trait and U is its generic point, every semi-abelian prolongation P of A coincides with the identity component N^0 of the Néron model N of A : since P is smooth with connected fibres, there is a morphism $P \rightarrow N^0$ inducing the identity on the generic fibre, and SGA 7 IX, 3.1 shows that it is an isomorphism. For normal noetherian S , two prolongations coincide on an open $V \supset U$ whose complement has codimension at least two. Raynaud's extension theorem then gives the result. A correction to this argument is supplied in the note at the end of the exposé.

1.3

Remark 1.2. *If P and Q are two semi-abelian algebraic spaces over S , with S normal and integral and with generic fibres P_η and Q_η abelian varieties over $k(\eta)$, the same argument gives*

$$\mathrm{Hom}_S(P, Q) \hookrightarrow \mathrm{Hom}_\eta(P_\eta, Q_\eta).$$

Remark 1.3. *The normality hypothesis on S is indispensable. Let S be a nodal cubic over an algebraically closed field, obtained from $\mathbf{P}^1$ by identifying 0 and ∞ in one point p ; let $U = S - \{p\}$; and let A be the constant elliptic curve E over U . To give a prolongation of A is to give a prolongation over $\mathbf{P}^1$ and a gluing isomorphism between the fibres at 0 and at ∞ . The prolongation over $\mathbf{P}^1$ is unique, namely $E \times \mathbf{P}^1$, but the gluing isomorphism can be chosen in at least two ways because $\mathrm{Aut}(E)$ contains $\{\pm 1\}$.*

The theorem should nevertheless remain true when U is schematically dense in S and S is only assumed geometrically unibranch. For general S , two prolongations inducing the same prolongations after the relevant normalizations should be isomorphic.

1.4 To avoid references that are not available, and because this case suffices for the intended application, we state Gabber's lemma under hypotheses that are probably unnecessarily restrictive.

Theorem 1.4 (Gabber's lemma). *Let U be a dense open subscheme of an excellent normal integral scheme S , and let A be an abelian algebraic space over U . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S' .

1.5 If there existed a moduli space for semi-abelian algebraic spaces, the semi-stable reduction theorem together with the valuative criterion of properness would show that this moduli object is proper over $\mathrm{Spec}(\mathbf{Z})$, and Gabber's lemma would follow immediately. But we are far from such a situation: the objects considered have automorphisms, sometimes infinitely many, and worse, a semi-abelian algebraic space over a complete discrete valuation ring is not determined by the associated formal algebraic space. If the special fibre is isomorphic to $\mathbf{G}_m$, the reduction modulo $\mathfrak{m}^k$ is isomorphic to that of $\mathbf{G}_m$ for every k by rigidity of tori, without A necessarily being a torus.

The bad behaviour of semi-abelian spaces with respect to formal completion comes from the fact that they are not necessarily proper. For curves the situation is better because one has the notion of a stable curve of genus g . Such curves are proper and, for $g \geq 2$, have only finitely many automorphisms. Because of finite automorphism groups, stable curves of genus $g \geq 2$ do not give a fine moduli space, but they do give a proper algebraic stack over $\mathrm{Spec}(\mathbf{Z})$ by the valuative criterion of properness and the stable reduction theorem for curves.

1.6

Lemma 1.5. *Let U be a quasi-compact dense open subspace of an algebraic space S , and let $X \rightarrow S$ be a family of stable curves of genus $g \geq 2$ parametrized by U . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of X over $f^{-1}(U)$ extends to a family of stable curves over S' .

Proof. Fix $g \geq 2$, and for brevity say simply “stable curve” for stable curve of genus g . Let $\mathcal{M}$ be the algebraic stack classifying stable curves. It is proper over $\mathrm{Spec}(\mathbf{Z})$. There is therefore a proper scheme S_0 over $\mathrm{Spec}(\mathbf{Z})$ and a surjective morphism $S_0 \rightarrow \mathcal{M}$; equivalently, S_0 is endowed with a family of stable curves $u_0 : X_0 \rightarrow S_0$ such that every stable curve over an algebraically closed field occurs as a fibre of X_0/S_0 .

Over $U \times S_0$ we have the two stable curves $\mathrm{pr}_1^* X$ and $\mathrm{pr}_2^* X_0$. Let

$$I = \mathrm{Isom}(\mathrm{pr}_1^* X, \mathrm{pr}_2^* X_0).$$

This algebraic space is finite over $U \times S_0$ and maps onto U . Let Γ be its schematic image in $U \times S_0$, and let $\bar{I}$ and $\bar{\Gamma}$ be the schematic closures inside $S \times S_0$:

$$\begin{array}{ccccc} I & \longrightarrow & \Gamma & \hookrightarrow & U \times S_0 \\ \downarrow & & \downarrow & & \downarrow \\ \bar{I} & \longrightarrow & \bar{\Gamma} & \hookrightarrow & S \times S_0. \end{array}$$

The algebraic space $\bar{I}$, endowed with the proper morphism

$$\bar{I} \longrightarrow \bar{\Gamma} \hookrightarrow S \times S_0 \longrightarrow S,$$

is the desired S' . The inverse image over $\bar{I}$ of the universal family X_0/S_0 extends the inverse image over I of X/U . $\square$

In Section 3 we shall demystify the use of algebraic stacks in this proof.

1.7 The principle of the proof of Gabber’s lemma is to reduce to the case of curves by using Jacobians. We record some preliminary remarks.

- (a) As explained in 1.1(a), A/U is automatically an abelian scheme.
- (b) Given $U \hookrightarrow S$, A/U , and $f : S' \rightarrow S$ as in Theorem 1.4, every proper surjective $f_1 : S'' \rightarrow S$ lying over S' again satisfies the conclusion. Applying Chow’s lemma and normalizing, if the theorem holds at all then it holds with $f : S' \rightarrow S$ projective and surjective, S' integral and normal.
- (c) It suffices to verify the theorem locally on S . If S is covered by finitely many open subspaces S_i , and over each S_i one has a solution $S'_i \rightarrow S_i$, one can find a projective surjective $S' \rightarrow S$, with $S'|_{S_i}$ lying over S'_i , with $f^{-1}(U)$ dense in S' , and with S' normal. The local prolongations glue by Theorem 1.1.
- (d) It suffices to verify the theorem after replacing U by a non-empty open subspace $V \subset U$: over integral normal S' , a prolongation of $A|_V$ is automatically a prolongation of A .

2 Proof of Gabber's lemma

Let $U \subset S$ and A/U be as in Theorem 1.4, and let d be the relative dimension of A over U . We may and shall assume $d \geq 1$. We seek an integral normal S' and a projective surjective morphism $f : S' \rightarrow S$ such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S' . In doing so we may shrink U and replace S by an integral normal scheme S_1 mapping projectively and surjectively to S , replace U by its inverse image, and replace A by its inverse image.

Let $\eta = \text{Spec } K$ be the generic point of U , and let A_η be the fibre of A/U at η . We may assume K infinite, otherwise $S = U = \text{Spec } K$. Choose a projective embedding

$$A_\eta \hookrightarrow \mathbf{P}^N.$$

In the Grassmannian of linear subspaces $H \subset \mathbf{P}^N$ of codimension $d - 1$, those for which $H \cap A_\eta$ is a smooth curve form the points of a dense open subset, hence have K -points. This density follows by iteration from the fact that a general hyperplane section is transverse. Choose such an H and put

$$T_\eta = A_\eta \cap H.$$

Over an algebraic closure $\overline{K}$ of K the weak Lefschetz theorem gives

$$H^0(A_{\overline{K}}, \mathbf{Z}_\ell) \longrightarrow H^0(T_{\overline{K}}, \mathbf{Z}_\ell), \quad H^1(A_{\overline{K}}, \mathbf{Z}_\ell) \hookrightarrow H^1(T_{\overline{K}}, \mathbf{Z}_\ell).$$

Thus $T_{\overline{K}}$ is connected and the morphism

$$\text{Alb}(T_\eta) = \text{Pic}^0(T_\eta) \longrightarrow A_\eta$$

from the Jacobian of T_η to A_η is an epimorphism. In particular T_η has genus $g \geq d$. Temporarily exclude the case $d = 1$; then $g \geq 2$.

After shrinking U there exists a smooth curve T_U of genus $g \geq 2$, embedded in A_U , such that

$$\text{Pic}^0(T_U) \twoheadrightarrow A_U.$$

Applying Lemma 1.5 to the curve T_U over $U \subset S$, and replacing S by the resulting modification (or by the normalization of one irreducible component), we may assume that T_U extends to a family of stable curves T over S .

By Artin's representability theorem, $\text{Pic}^0(T)$ exists as an algebraic space over S . Let

$$P := \text{Pic}^0(T)$$

be the open algebraic subspace in groups parametrizing invertible sheaves on T whose restriction to every irreducible component of every geometric fibre has degree zero. This algebraic space is separated and smooth of finite type over S . Its geometric fibres are extensions of abelian varieties by tori; hence P is a semi-abelian algebraic space over S .

We have therefore obtained, after the allowed modification of S and shrinking of U , a semi-abelian algebraic space P over S such that P_U is abelian and carries an epimorphism

$$u : P_U \rightarrow A.$$

Let $K_U = \text{Ker}(u)$. It is a subgroup algebraic subspace of P_U . After shrinking U if necessary, we may suppose it flat over U . By flattening by modification, a further modification of S permits us to suppose that the schematic closure K of K_U in P is flat over S ; then K is automatically a subgroup algebraic space of P .

For algebraic spaces in groups over a base S , the quotient by a flat closed subgroup algebraic space exists, is compatible with base change, is separated over S , and is flat if the original group space is flat. Let

$$N = P/K.$$

The geometric fibres of N are quotients of extensions of abelian varieties by tori, hence again extensions of abelian varieties by tori. Thus N is semi-abelian, and since $A = P_U/K_U$, it extends A and proves the theorem.

It remains only to handle the case $d = 1$. The same proof applies if one defines a stable curve of genus one as a pointed curve $X \rightarrow S$ with section in the smooth locus and with geometric fibres either elliptic curves or nodal cubics. Alternatively, one may define T_η as a hyperplane section of $A_\eta \times A_\eta$.

3 Exorcising stacks

3.1 Recall the definition of a groupoid in schemes. It is a diagram

$$S_2 \rightrightarrows S_1 \rightrightarrows S_0 \quad (3.1.1)$$

with face maps d_i , $i = 0, 1$ and $i = 0, 1, 2$ respectively, such that the square

$$\begin{array}{ccc} S_2 & \longrightarrow & S_1 \\ \downarrow & & \downarrow d_1 \\ S_1 & \xrightarrow{d_0} & S_0 \end{array}$$

is cartesian and such that, for every scheme T , the sets $\text{Hom}(T, S_i)$ define a groupoid. The objects are the elements of $\text{Hom}(T, S_0)$, the arrows are those of $\text{Hom}(T, S_1)$, the source and target of an arrow f are $d_1(f)$ and $d_0(f)$, and if the target of f is the source of g , the composition is $g \circ f = d_1\langle g, f \rangle$, where $\langle g, f \rangle \in \text{Hom}(T, S_2)$ is determined by $d_2\langle g, f \rangle = f$ and $d_0\langle g, f \rangle = g$. We denote this groupoid by $S_*(T)$. As T varies, the $S_*(T)$ form a prestack in groupoids.

The quasi-compact separated algebraic stacks are the stacks in groupoids obtained from such a prestack, for the étale topology, when S_0 is quasi-compact, $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$ is finite, and the $d_i : S_1 \rightarrow S_0$ are étale.

The quasi-compact separated algebraic spaces correspond to the particular case where $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$ is a closed immersion. Then $S_1 \subset S_0 \times S_0$ is an étale equivalence relation, and the algebraic space is the quotient S_0/S_1 .

3.2 Let $g \geq 2$, and say simply stable curve for stable curve of genus g . We explain how to construct a diagram (3.1.1) for the stack $\mathcal{M}$ of stable curves. If T_i is endowed with a stable curve $u_i : X_i \rightarrow T_i$ for $i = 1, 2$, write

$$T_1 \times_{\mathcal{M}} T_2 = \text{Isom}(X_1, X_2).$$

This represents the functor sending T to triples (a_1, a_2, φ) , where $a_i : T \rightarrow T_i$ and $\varphi : a_1^*X_1 \xrightarrow{\sim} a_2^*X_2$. This fibre product is finite over $T_1 \times T_2$.

The Hilbert scheme H of stable curves X endowed with a tricanonical embedding

$$X \hookrightarrow \mathbf{P}^{5g-6}$$

is smooth over $\mathcal{M}$ and maps onto $\mathcal{M}$. The group $\text{PGL}(5g-5)$ acts on H with finite reduced stabilizers. If S is a disjoint sum of smooth subschemes of H , transverse to the PGL -orbits, of complementary dimension and meeting all of them, then S is étale over $\mathcal{M}$ and maps onto it, hence supplies a diagram (3.1.1). The stack $\mathcal{M}$ is separated of finite type over $\text{Spec}(\mathbf{Z})$. Since deformations of stable curves are unobstructed, it is smooth; this also follows from the smoothness of H over $\text{Spec}(\mathbf{Z})$.

3.3 Let $n \geq 1$. Let $\mathcal{M}^0[n]$ be the algebraic stack classifying proper smooth curves X of genus g over bases S where n is invertible, together with a level- n structure

$$\alpha : \text{Pic}^0(X)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

If $\mathcal{M}^0$ is the open part of $\mathcal{M}$ corresponding to smooth curves, the forgetful morphism

$$\mathcal{M}^0[n] \longrightarrow \mathcal{M}^0[1/n]$$

makes $\mathcal{M}^0[n]$ a finite étale cover of $\mathcal{M}^0[1/n]$.

3.4

Definition 3.1. Let $\mathcal{M}_n$ be the normalization of $\mathcal{M}$ in $\mathcal{M}^0[n]$.

In the language of 3.1 this means the following. Choose S_0 étale over $\mathcal{M}$ and mapping onto it, whence a diagram S_* . Over each S_i there is a stable curve X_i , and these form a cartesian system

$$\begin{array}{ccccc} X_2 & \rightrightarrows & X_1 & \rightrightarrows & X_0 \\ \uparrow & & \uparrow & & \uparrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0. \end{array}$$

Let S_i^0 be the open subset over which X_i is smooth, and let $S_{i,n}[1/n]$ be the finite étale cover of $S_i^0[1/n]$ given by

$$\text{Isom}(\text{Pic}^0(X_i)_n, (\mathbf{Z}/n\mathbf{Z})^{2g}).$$

Let $S_{i,n}$ be the normalization of S_i in $S_{i,n}[1/n]$. Since normalization commutes with étale base change in this situation, the $S_{i,n}$ again form a diagram (3.1.1), defining $\mathcal{M}_n$. The diagram

$$\begin{array}{ccccc} S_{2,n} & \rightrightarrows & S_{1,n} & \rightrightarrows & S_{0,n} \\ \downarrow & & \downarrow & & \downarrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0 \end{array}$$

is cartesian. In particular, for $t \in S_{0,n}(T)$ with image $s \in S_0(T)$ one has an injection of automorphism groups

$$\text{Aut}(t) \hookrightarrow \text{Aut}(s). \tag{3.4.1}$$

3.5

Proposition 3.2. If $n \geq 3$, then $\mathcal{M}_n[1/n]$ is an algebraic space.

The essential point is the following lemma.

Lemma 3.3. Let X be a stable curve over an algebraically closed field k , and let $n \geq 3$ be invertible in k . Every automorphism of X acting trivially on $\text{Pic}^0(X)_n$ is the identity.

Proof. Write $\text{Pic}^0(X)$ as an extension of an abelian variety by a torus,

$$0 \longrightarrow T \longrightarrow \text{Pic}^0(X) \longrightarrow A \longrightarrow 0,$$

and let L be the character group of T . Every automorphism u of X has finite order. If it acts trivially on $\text{Pic}^0(X)_n$, it acts trivially on A_n and on T_n , hence on L/nL . By Serre's rigidity for A , and by the analogous direct argument for L , it acts trivially on A and on L , hence on T . Thus $u - 1$ sends $\text{Pic}^0(X)$ to T and is therefore zero; u acts trivially on $\text{Pic}^0(X)$, and the conclusion follows from the standard faithfulness result for stable curves. $\square$

Lemma 3.4. *Let S be normal, let $j : U \hookrightarrow S$ be a dense open subset, let A be an abelian algebraic space over U , let N be a semi-abelian prolongation of A over S , and let n be invertible on S .*

- (i) *The étale sheaf N_n of points killed by multiplication by n injects into j_*A_n .*
- (ii) *Every isomorphism over U from A_n to the constant sheaf $(\mathbf{Z}/n\mathbf{Z})^{2g}$ extends to a monomorphism*

$$N_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

Proof. The first assertion follows from the separatedness of N_n over S . Since S is normal, the constant sheaf satisfies

$$(\mathbf{Z}/n\mathbf{Z})^{2g} \simeq j_*(\mathbf{Z}/n\mathbf{Z})^{2g},$$

and the composite

$$N_n \longrightarrow j_*j^*N_n = j_*A_n \longrightarrow j_*(\mathbf{Z}/n\mathbf{Z})^{2g} = (\mathbf{Z}/n\mathbf{Z})^{2g}$$

is the required prolongation. □

Proof of the proposition. Let $T_1 = S_{1,n}[1/n]$. Since T_1 is finite and unramified over $T_0 \times T_0$, it suffices to verify that for every algebraically closed field k , the map

$$T_1(k) \longrightarrow T_0(k) \times T_0(k)$$

is injective. We may suppose n invertible in k , otherwise $T_1(k) = \emptyset$. Because $T_*(k)$ is a groupoid, it suffices to verify that for $t \in T_0(k)$ there is exactly one object of $T_1(k)$ over (t, t) , i.e. that $\text{Aut}(t)$ is reduced to the identity.

Let X denote the stable curve obtained by pullback of the universal curve. On $T_i^0 = S_{i,n}[1/n]$ there is by definition a cartesian system of isomorphisms

$$\text{Pic}^0(X_i)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

By Lemma 3.4 it extends over T_i to a monomorphism of étale sheaves. It follows that the morphism from the stack defined by T_* to that defined by S_* factors through the stack of stable curves X endowed with a monomorphism $\text{Pic}^0(X)_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}$. For $t \in T_0(k)$, the injection (3.4.1) has image consisting of automorphisms of the corresponding stable curve acting trivially on $\text{Pic}^0(X)_n$. By Lemma 3.3, such an automorphism is trivial. □

3.6

Corollary 3.5. *If n is divisible by two integers n_i , $i = 1, 2$, prime to one another and each at least 3, then $\mathcal{M}_n$ is an algebraic space.*

It suffices to verify that each $\mathcal{M}_n[1/n_i]$ is an algebraic space, i.e. that the corresponding objects have no non-trivial automorphisms. This follows from (3.4.1) and its factorization

$$\mathcal{M}_n[1/n_i] \longrightarrow \mathcal{M}_{n_i}[1/n_i] \longrightarrow \mathcal{M},$$

together with Proposition 3.5.

3.7 In 3.6, stack language gave an étale atlas defining an algebraic space $\mathcal{M}_n$. The valuative criteria of separatedness and properness, together with the semi-stable reduction theorem, show that this algebraic space is proper over $\text{Spec}(\mathbf{Z})$. One may use this $\mathcal{M}_n$ as the S_0 in the proof of Lemma 1.5. If one wants S_0 to be projective over $\text{Spec}(\mathbf{Z})$, one applies Chow's lemma to $\mathcal{M}_n$. In fact $\mathcal{M}_n$ is projective, by Mumford.

4 Obtaining schemes

In this section we show how to modify and complete the proof of Gabber's lemma so as to obtain schemes rather than algebraic spaces. Another approach was indicated to me by Michel Raynaud: he says he can prove that if X is an S -algebraic space in groups, smooth and separated over S , with connected fibres, then X becomes representable after base change to a suitable blow-up S'/S .

Let $g \geq 2$. For brevity again say stable curve for stable curve of genus g .

4.1 For a smooth curve X/S of genus g , autoduality of the Jacobian

$$\varepsilon : \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(X)^*$$

gives a canonical relatively ample invertible sheaf on $\mathrm{Pic}^0(X)$, namely

$$(1, \varepsilon)^*(\text{Poincaré sheaf}).$$

It is symmetric and is trivialized along the zero section. Its formation is compatible with arbitrary base change. Denote it by $\mathcal{L}(X/S)$.

4.2

Lemma 4.1. *It is possible, uniquely up to unique isomorphism, to attach to each stable curve X/S an invertible sheaf $\mathcal{L}(X/S)$ on $\mathrm{Pic}^0(X)$, trivialized along the zero section, compatible with all base change, and equal to the sheaf of 4.1 when X is smooth.*

Proof. It suffices to treat the universal case. With the notation of 3.4, choose X_*/S_* . We seek invertible sheaves $\mathcal{L}(X_i/S_i)$ on the $\mathrm{Pic}^0(X_i)$, trivialized along the zero section, extending the sheaves of 4.1 over the open subset where X_i is smooth and forming a cartesian system under the maps $S_i \rightarrow S_j$. Each S_i is smooth over $\mathrm{Spec}(\mathbf{Z})$, hence regular, and so is $\mathrm{Pic}^0(X_i)$, which is smooth over S_i . An extension exists by taking the determinant of any coherent extension. Correcting by the inverse image of its restriction to the zero section gives a trivialized extension. Since $\mathrm{Pic}^0(X_i)$ has geometrically irreducible fibres, such an extension is unique up to unique isomorphism, and these unique isomorphisms give the required compatibility. $\square$

4.3

Proposition 4.2. *For a stable curve X/S , the invertible sheaf $\mathcal{L}(X/S)$ of 4.2 is relatively ample. In particular, $\mathrm{Pic}^0(X)$ is quasi-projective over S and is a scheme if S is a scheme.*

As before it suffices to verify this in a universal case; one may suppose S regular and the open subset $S^0 \subset S$ over which X is smooth dense. If one already knew that $\mathrm{Pic}^0(X)$ were a scheme, the assertion would follow from Raynaud's theorem. Unfortunately that proof uses the scheme hypothesis essentially in order to construct, locally on S , relatively ample divisors as complements of affine open subsets.

4.4

Lemma 4.3. *Let X be a geometrically reduced connected curve over a separably closed field k . There exists a divisor D , contained in the smooth locus of X , of degree*

$$g = \dim H^1(X, \mathcal{O}_X),$$

such that

$$H^0(X, \mathcal{O}_X(D)) = k, \quad H^1(X, \mathcal{O}_X(D)) = 0.$$

Proof. The assertion on H^0 follows from the other assertion by Riemann-Roch. If ω is the dualizing sheaf, then $\dim H^0(X, \omega) = g$, and by duality $H^1(X, \mathcal{O}_X(D)) = 0$ is equivalent to $H^0(X, \omega(-D)) = 0$. The sheaf ω has no embedded component. We may therefore construct by induction divisors D_i of degree i , contained in the smooth locus, with

$$\dim H^0(X, \omega(-D_i)) = g - i,$$

by taking $D_0 = 0$ and $D_{i+1} = D_i + P_i$, where P_i is a smooth k -point at which not all sections of $H^0(X, \omega(-D_i))$ vanish. Take $D = D_g$. $\square$

4.5

Lemma 4.4. *Let $f : X \rightarrow S$ be projective, with geometrically reduced connected fibres. Locally for the étale topology on S , there is an open subset*

$$V \subset \mathrm{Pic}^0(X)$$

meeting every fibre and which is a scheme.

Proof. One reduces to the case where S is strictly local with closed point s , and it suffices to construct an open $V \subset \mathrm{Pic}^0(X)$ which is a scheme and meets the fibre over s . By Lemma 4.3, choose a divisor D_s of degree $g = \dim H^1(X_s, \mathcal{O}_{X_s})$ on X_s , contained in the smooth locus, and satisfying $H^1(X_s, \mathcal{O}(D_s)) = 0$. Extend it to a relative Cartier divisor D on X , possible because S is local henselian. Then

$$f_*\mathcal{O}(D) = \mathcal{O}_S, \quad R^1f_*\mathcal{O}(D) = 0.$$

Let V' be the open subset of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves $\mathcal{F}$ such that $R^1f_*\mathcal{F}(D) = 0$. For such $\mathcal{F}$, $f_*\mathcal{F}(D)$ is free of rank one and commutes with all base change. Let $V \subset V'$ be the open corresponding to those $\mathcal{F}$ for which the zero locus of a basis section of $f_*\mathcal{F}(D)$ is a relative Cartier divisor $D(\mathcal{F})$ contained in the smooth locus X^0 . By construction V contains the zero section. The map

$$\mathcal{F} \longmapsto D(\mathcal{F})$$

identifies V with an open subset of $\mathrm{Sym}^g(X^0)$, and hence V is a scheme. $\square$

4.6

Lemma 4.5. *Under the hypotheses of Lemma 4.4, $\mathrm{Pic}^0(X)$ is locally a scheme for the étale topology on S .*

Proof. Again suppose S strictly local. Put $P = \mathrm{Pic}^0(X)$. It is a smooth algebraic space over S , pure of relative dimension g . Let V be as in Lemma 4.4. It suffices to show that P is the union of the translates gV for $g \in P(S)$. Reducing to the case where S is essentially of finite type over $\mathbf{Z}$, one may count dimensions. Let P^n be the n -fold fibre product of P over S . For a geometric point t of S and a geometric point $\gamma = (\gamma_1, \dots, \gamma_n)$ of P^n over t , say that γ is allowed if the translates $\gamma_i V$ cover P_t . The allowed points form a constructible subset Y of P^n . If Z is its complement, then fibre by fibre Z has codimension at least $n - g$. For $n - g > \dim S$, the closure of Z cannot contain the whole special fibre, so one finds a section γ of $P^n - \overline{Z}$, and then P is the union of the $\gamma_i V$. $\square$

4.7 In Lemmas 4.4 and 4.5, the projectivity hypothesis on f may be replaced by properness: locally on S for the étale topology, f is automatically projective.

4.8

Proof of Proposition 4.2. The problem is local on S for the étale topology. By Lemma 4.5, one can apply Raynaud's theorem giving relative ampleness in the scheme case. $\square$

4.9 The proof of Raynaud's theorem seems to use the scheme hypothesis only through the conclusion of Lemma 4.4. This would allow one to avoid Lemma 4.5.

4.10

Proposition 4.6. *Gabber's lemma remains true with “semi-abelian algebraic space” in the conclusion replaced by “semi-abelian scheme.”*

Proof. Proceeding as in Section 2, one reduces to the case where there is a stable curve X over S , smooth over U , and an epimorphism over U

$$u : \mathrm{Pic}^0(X) \longrightarrow A$$

whose kernel K_U extends to a flat subgroup algebraic space K of $\mathrm{Pic}^0(X)$. It remains to show that

$$P := \mathrm{Pic}^0(X)/K$$

is a scheme.

Over U , Poincaré complete reducibility gives a morphism

$$v : A \longrightarrow \mathrm{Pic}^0(X)$$

such that uv is multiplication by an integer $n \geq 1$ on A . By the extension result of 1.3, v extends to a morphism, again denoted v , from P to $\mathrm{Pic}^0(X)$. By density of U in S , one still has $uv = n$. Since multiplication by n on P is quasi-finite, v is quasi-finite, hence quasi-affine. Since $\mathrm{Pic}^0(X)$ is a scheme, P is a scheme. $\square$

I thank M. Raynaud and O. Gabber for a critical reading of a first version of this text.

Note

(1).1 Let $S = \mathrm{Spec} A$ be affine, and let $U \subset S$ be an open subscheme which is a finite union of principal opens $U_i = \mathrm{Spec} A[f_i^{-1}]$. If F is a closed subscheme of U , defined in U_i by an ideal $I_i \subset A[f_i^{-1}]$, then the schematic closure $\overline{F}$ of F in S is the closed subscheme defined by

$$I = \bigcap_i \rho_i^{-1}(I_i),$$

where $\rho_i : A \rightarrow A[f_i^{-1}]$ is the canonical map. If $S' = \mathrm{Spec} A'$ is flat over S , then the inverse image of $\overline{F}$ in S' is again the schematic closure of the inverse image of F . Hence schematic closure commutes with étale localization and so makes sense for algebraic spaces.

More generally, for $f : X \rightarrow S$ quasi-compact, the closed schematic image $f(X)^-$ is the closed subscheme defined by the ideal of elements $a \in A$ whose inverse image on X is zero. For f the inclusion of F as above, this recovers schematic closure. The same base-change compatibility holds for schematic images.

(1).2 Let S be a noetherian algebraic space, and let X and Y be two algebraic spaces over S , each endowed with a section e . Let $V \subset X$ be open, put $U = e^{-1}(V) \subset S$, and let

$$f_V : V \longrightarrow Y$$

be an S -morphism sending e to e . For each n , f_V induces a morphism from the n -th infinitesimal neighbourhood of e in V to the corresponding neighbourhood in Y_U . Passing to the inverse limit gives a morphism

$$f_V^\wedge : X_U^\wedge \longrightarrow Y_U^\wedge$$

between formal completions along the sections.

Proposition 4.7. *Assume X smooth over S and Y separated of finite type.*

- (i) *If U contains all points of S of depth at most 1, then $f_V^\wedge$ extends uniquely to a morphism $f^\wedge : X^\wedge \rightarrow Y^\wedge$ between the formal completions of X and Y along the sections.*
- (ii) *Let Γ be the schematic closure in $X \times_S Y$ of the graph $\Gamma_V \subset V \times_S Y$ of f_V . If $f_V^\wedge$ extends to $f^\wedge : X^\wedge \rightarrow Y^\wedge$ and U contains all points of S of depth 0, then Γ is étale over X along the section e , and its formal completion there is the graph of $f^\wedge$.*

The proof is local on S for the étale topology; thus one may suppose S affine. Then the infinitesimal neighbourhoods of the section in X and Y are affine schemes, and (i) is proved as in Raynaud. Smoothness of X/S ensures that the infinitesimal neighbourhoods of e in X are finite and flat over S .

We use the following lemma for (ii).

Lemma 4.8. *Let $S = \operatorname{Spec} A$ with A a complete noetherian local ring, let*

$$X = \operatorname{Spec} R[[T_1, \dots, T_n]],$$

let e be the zero section, let V be an open subset of X such that $U = e^{-1}(V)$ contains all associated points of S , and let $F \subset V$ be a closed subscheme containing all infinitesimal neighbourhoods of $e(U)$ in V . Then $F = V$.

Proof. It suffices to show that X is the schematic closure of F in X . If

$$f = \sum a_I T^I$$

vanishes on F , then each coefficient a_I vanishes on $U \subset S$, hence is zero, and $f = 0$. $\square$

Proof of Proposition 4.7(ii). Let s be a point of S . Write $X_s^\wedge$ for the completion of X at $e(s)$, and let $V_s^\wedge$ and U_s be the inverse images of V and U . The morphism $f^\wedge$ induces $X_s^\wedge \rightarrow Y$, and f_V induces $V_s^\wedge \rightarrow Y$. These two morphisms coincide on a closed subscheme containing all infinitesimal neighbourhoods of $e(U_s)$, hence coincide by Lemma 4.8. The inverse image of Γ_V in $X_s^\wedge \times_S Y$ is therefore the graph of $f^\wedge|_{V_s^\wedge}$, whose schematic closure is the graph of $f^\wedge : X_s^\wedge \rightarrow Y$. Since schematic closure commutes with the flat base change $X_s^\wedge \times_S Y \rightarrow X \times_S Y$, it follows that after the base change $X_s^\wedge \rightarrow X$, the space Γ is étale over X . Hence Γ is étale over X at $e(s)$. $\square$

(1).3

Corollary 4.9. *Let G and H be algebraic spaces in groups over S , let U be an open subspace of S , and let*

$$f_U : G_U \longrightarrow H_U$$

be a morphism. Assume that G is smooth over S with connected fibres, that H is of finite type and separated over S , and that U contains all points of S of depth 0. If the formal completion of f_U along the zero sections extends to a morphism of formal completions

$$G^\wedge \longrightarrow H^\wedge,$$

then f_U extends uniquely to a morphism $G \rightarrow H$.

If U contains all points of depth at most 1, the existence of the formal extension is guaranteed by Proposition 4.7(i).

Proof. Uniqueness follows from the fact that U is schematically dense in S , hence G_U is schematically dense in G . For existence, let Γ_U be the graph of f_U , let $\Gamma \subset G \times_S H$ be its schematic closure, and let Γ^0 be the open subset of Γ where Γ is étale over G . By Proposition 4.7(ii), Γ^0 contains the zero section of $G \times_S H$. For every section γ of Γ , left translation by the element of $G_U \times_S H_U$ determined by $\gamma|_U$ sends Γ_U to itself. By transport of structure, left translation by γ sends Γ^0 to itself. After the flat base change Γ^0/S , this shows that Γ^0 is a subgroup algebraic space of $G \times_S H$.

The image of Γ^0 in G is an open subgroup algebraic space, hence is all of G because G/S has connected fibres. Since Γ^0/G is étale and separated, the function assigning to x the number of geometric points in the fibre over x is lower semicontinuous. It is at least one everywhere and equals one over G_U , hence is constantly one. Therefore $\Gamma^0 \rightarrow G$ is étale, surjective, and radicial, hence an isomorphism. The required extension is the composite

$$G \xleftarrow{\sim} \Gamma^0 \longrightarrow H.$$

□

(1).4 The preceding proposition should remain true if the hypothesis that G/S is smooth is weakened to flatness of finite type, and if the hypothesis that H is separated over S is omitted.

Bibliography

- [1] M. Artin, *The implicit function theorem in algebraic geometry*, in Proc. Colloquium on Algebraic Geometry, Tata Institute, 1969.
- [2] M. Artin, *Algebraization of formal moduli I*, in *Global Analysis, Papers in Honor of K. Kodaira*, Princeton University Press, 1969, pp. 21–71.
- [3] M. Artin and G. Winters, *Degenerate fibres and stable reduction of curves*, Topology 10 (1971), 373–383.
- [4] W. L. Chow, *On the projective embedding of homogeneous varieties*, in a symposium in honor of S. Lefschetz, Princeton University Press, pp. 122–128.
- [5] P. Deligne and D. Mumford, *The irreducibility of the space of curves of a given genus*, Publ. Math. IHES 36 (1969), 75–110.
- [6] D. Knutson, *Algebraic Spaces*, Lecture Notes in Math. 203, Springer, 1971.
- [7] J.-P. Murre, *Representation of unramified functors. Applications*, Séminaire Bourbaki 294, 1964–1965.
- [8] M. Raynaud, *Modèles de Néron*, C. R. Acad. Sci. 262 (1966), 413–416.
- [9] M. Raynaud, *Spécialisation du foncteur de Picard*, Publ. Math. IHES 38 (1970), 27–76.
- [10] M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes in Math. 119, Springer, 1970.
- [11] M. Raynaud and L. Gruson, *Critères de platitude et de projectivité*, Invent. Math. 13 (1971), 1–39.
- [12] J.-P. Serre, *Rigidité du foncteur de Jacobi d'échelon $n \geq 3$* , appendix to Exposé 17 of Séminaire Cartan 1960/61.

Sigla

EGA denotes *Éléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné. SGA denotes *Séminaire de géométrie algébrique du Bois-Marie*.

Heisenberg Group and Reality

P. Deligne

1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ) , the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$\tau : (x, y, \lambda) \mapsto (-x, -y, \lambda) \quad (1.2)$$

is an automorphism of H .

The coordinate system $x, y, \lambda : H \rightarrow \mathbb{C}^2 \times \mathbb{C}^*$ exhibits the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section, $\tau(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, by means of this section, a lift to a one-parameter subgroup of H , again denoted by D . The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$, with

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H , and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which as a set is identified with $\mathbb{R}^2 \times U^1$.

It is known, by the Stone-von Neumann theorem [2], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V for which $\lambda \in U^1$ acts by multiplication by λ . In the Schrödinger model of V , one has $V = L^2(\mathbb{R})$, and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (x, 0, 1) : f(z) &\mapsto f(z + x), \\ (0, y, 1) : f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of V one deduces a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V , compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [3], D. Shale [1]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathcal{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V , are the L^2 functions on $\mathbb{R}$ that extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\mathrm{Im} z| < A_1$, one has, for all $A_2 \in \mathbb{R}$,

$$|f(z)| \leq O(\exp(-A_2|z|)).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol} , one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V . If one did, the projective action of $\mathrm{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\mathrm{SL}(2, \mathbb{C})$ on V^{hol} ; since $\mathrm{SL}(2, \mathbb{C})$ is simply connected, this action would lift to an honest action of $\mathrm{SL}(2, \mathbb{C})$, and the projective action of $\mathrm{SL}(2, \mathbb{R})$ would lift to an honest action. This is not the case: only the double cover of $\mathrm{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

2

Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines of $\mathbb{C}^2$. For $D \in S$, lifted in H as in no. 1, let W_D be the space of holomorphic functions φ on H satisfying

$$\varphi(\lambda dh) = \lambda \varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by $g : \varphi(h) \mapsto \varphi(hg)$. For $E \neq D$, restriction to E is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \quad (2.1)$$

where $\mathcal{O}(E)$ denotes the space of holomorphic functions on E .

If R is a holomorphic representation of H , on which each $\lambda \in \mathbb{C}^*$ acts by λ , and if ω is a D -invariant linear form on R , then

$$w \mapsto \omega(hw) \quad (2.2)$$

is a morphism from R to W_D .

3. Example

Let V , as in nos. 1 and 2, be the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathcal{S} \subset V$ be the space of its C^∞ vectors, and let $\mathcal{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\text{Im } t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t : f \mapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t) dx$$

is defined on V , and even on $\mathcal{S}'$. The form ω_t is a holomorphic vector of the dual of V , fixed by D_t . In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t , the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2), ω_t defines

$$\Omega_t : V^{\text{hol}} \longrightarrow W_{D_t},$$

which extends to V and even to $\mathcal{S}'$. Let E be the x -axis. The map $r_E \Omega_t : V \rightarrow \mathcal{O}(E)$ sends $f \in V$ to

$$f(x, t) = r_E \Omega_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\left(\partial_t - \frac{1}{4\pi i} \partial_x^2 \right) f(x, t) = 0.$$

For real t , the form ω_t remains defined on $\mathcal{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $f \mapsto \int f(x) dx$. The homomorphism Ω_t remains defined on V^{hol} . For $t = 0$, $r_E \Omega_0$ is, in the Schrödinger model, the identity.

Assume $\text{Im } t > 0$. Let $\overline{D}_t$ be the complex conjugate of D_t . In the affine line $S - \{\overline{D}_t\}$, the “half-plane” $\{D_u \mid \text{Im } u > 0\}$ is a disk with center D_t . The map $r_{\overline{D}_t} \circ \Omega_t : V \rightarrow W_{D_t} \rightarrow \mathcal{O}(\overline{D}_t)$ gives the holomorphic model of V : the action of H is given by

$$\begin{aligned} d \in D_t : \quad & f(x) \mapsto f(x + d), \\ e \in \overline{D}_t : \quad & f(x) \mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned}$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d)\overline{f(\bar{d})}\exp(2\pi i\langle d, \bar{d} \rangle)$$

is invariant under $H_{\mathbb{R}}$; moreover $r_{\bar{D}_t} \circ \Omega_t$ identifies V with the space of f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i\langle d, \bar{d} \rangle)$, the commutator of d and $\bar{d}$, is

$$\exp(2\pi i\langle d, \bar{d} \rangle) = \exp(2\pi i(t - \bar{t})|z|^2).$$

4

Fix E . The isomorphisms r_E of (2.1) make it possible to regard the W_D for $D \neq E$ as a family of representations of H operating in the fixed space $\mathcal{O}(E)$, with the action of H depending holomorphically on D . Let us make explicit the dependence on E of r_E . Choose coordinates as in no. 1, with E the x -axis. Let E' be generated by $(1, u)$ and D by $(-t, 1)$. We use x as coordinate on both E and E' . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi i u x x')$. Hence $r_{E'} r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$, with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'}\left(\frac{x}{1+tu}\right) = \exp\left(\frac{\pi i u}{1+tu}x^2\right) f_E(x). \quad (4.1)$$

This formula is holomorphic in t : the W_D form a holomorphic vector bundle $\mathcal{W}$ of infinite dimension over S , on which H acts.

To understand (4.1), one may observe that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \neq D$ and of the coordinate z on E :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O(\exp(A|z|^2)). \quad (4.2)$$

Let $W_D^0 \subset W_D$ be defined by this condition. The W_D^0 form a holomorphic sub-bundle $\mathcal{W}^0$ of the bundle $\mathcal{W}$ in holomorphic representations of H .

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f :

- (i)_T There exists a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$, with initial condition $g(z, 0) = f(z)$.
- (ii)_T $|f(z)| \leq O(\exp(|z|^2/4T))$.
- (iii)_T $\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|dz \wedge d\bar{z}| < \infty$.

If $T < T_1$, each of $(i)_{T_1}$, $(ii)_{T_1}$, $(iii)_{T_1}$ implies each of $(i)_T$, $(ii)_T$, $(iii)_T$.

Proof. $(i)_{T_1} \Rightarrow (ii)_T$. The heat equation remains satisfied by the derivatives of g , and therefore

$$\partial_z^{2n} g = \partial_t^n g, \quad \partial_z^{2n+1} g = \partial_t^n (\partial_z g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \quad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for $\partial_z^{2n+1}g$. The Taylor series of f then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!} \frac{1}{T_2^n} |z|^{2n}\right).$$

One has

$$\frac{1}{2n+1} 2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n} \frac{1}{T^n} \frac{|z|^{2n}}{n!}\right),$$

and this sum is $\exp(|z|^2/4T)$.

$(ii)_{T_1} \Rightarrow (i)_T$. It suffices to take g given by the kernel of the heat equation:

$$g(z, t) = \int_{\sqrt{t}\mathbb{R}} f(z+x) \frac{1}{2\sqrt{\pi t}} \exp(-x^2/4t) dx.$$

Changing $\sqrt{t}$ to $-\sqrt{t}$ changes the orientation of the cycle $\sqrt{t}\mathbb{R}$ and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: $(ii)_{T_1} \Rightarrow (iii)_T$, and $(iii)_{T_1} \Rightarrow (ii)_T$ is verified by writing $f(z)$ as the mean of its values on a disk with center z and radius $1/|z|$, and estimating that mean by Cauchy–Schwarz.

6

In homage to the Stone-von Neumann theorem, the bundle $\mathcal{W}$ over S is, projectively, equipped with a holomorphic connection for which the action of H is horizontal.

Let us specify “projectively.” On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L endowed with an isomorphism $L^{\otimes 2} \simeq \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity ± 1 . This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathcal{W}$, but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \quad (6.1)$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and if ∇ is a connection on $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$, then its transport to $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$ of square roots of $\mathcal{O}(-1)$ does not depend on the choice of a , which can be changed locally only by a constant ± 1 . Thus the notion of a “connection on $\mathcal{W}(-\frac{1}{2})$ ” is well defined independently of the global existence, or the choice, of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D . For a local section s of $\mathcal{W}(-\frac{1}{2})$, the problem is to define $\nabla_t s$ in the fiber of $\mathcal{W}(-\frac{1}{2})$ at D . We proceed as follows: after making an auxiliary choice u , one defines $\nabla_t^u s$ for s a local section of $\mathcal{W}$ or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u , with dependence in u of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where a takes opposite values for $\mathcal{W}$ and for $\mathcal{O}(-\frac{1}{2})$. Hence, for $s = s_1 s_2$, with s_1 a local section of $\mathcal{W}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u .

Let $D \in S$ be a line of $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t : D \rightarrow \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D)^{\otimes 2}$: to $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$ one associates $t(d) = \langle u, d \rangle v'$, where u is any representative of u' . The auxiliary datum will be a lift $\tilde{t} : D \rightarrow \mathbb{C}^2$ of t , or equivalently a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it $\tilde{t} = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let δ_z be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of H . For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y \rangle \delta_{(0,0,1)}. \quad (6.2)$$

Let $\mathcal{U}_1$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta_{(0,0,1)} = 1$. In $\mathcal{U}_1$, (6.2) implies that, for $d \in D$,

$$\left[-\frac{\delta_u^2}{4\pi i}, \delta_d \right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d \rangle \delta_u = \langle u, d \rangle \delta_u = \delta_{t(d)}. \quad (6.3)$$

Let us compute to first order around D . For $\varepsilon^2 = 0$ and $\varphi(g)$ in W_D , so $\delta_d \varphi = 0$, one has

$$\delta_{d+\varepsilon t(d)} \left[\varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = \varepsilon \left[\delta_{t(d)} \varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = 0,$$

because $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$, by (6.3). To first order around D , the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle $\mathcal{W}$. If $\varphi(g, D')$ is a local section of $\mathcal{W}$, this permits one to define the derivative $\nabla_t^u \varphi$ in $\mathcal{W}_D$ by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on u . If u is replaced by $u + x$ with $x \in D$, then in $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u \delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u \delta_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For $\varphi \in W_D$, $\delta_x \varphi = 0$, and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

whence

$$\frac{1}{4\pi i} \delta_{u+x}^2 \varphi = \frac{1}{4\pi i} \delta_u^2 \varphi + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of u also makes it possible to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d \rangle u$ to obtain a homogeneous function of degree 1 on D . When u is changed into $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2} \nabla_t^u(s) s^{-3/2},$$

and, for the dependence on u , s now being a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x \rangle s. \quad (6.9)$$

The dependences on u in (6.8) and (6.9) cancel to provide a derivation ∇_t for $\mathcal{W}(-\frac{1}{2})$. Since the connection ∇ is defined in terms of right-invariant vector fields on H , it commutes with the action of H .

7

Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let E be the x -axis. On S , $\mathcal{O}(1)$ has a section s with a simple zero at E . It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1} . By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{W}$. We calculate it.

Let D_t be the line generated by $(-t, 1)$. By r_E of (2.1), a local section of $\mathcal{W}$ is identified with a function $f(z, t)$ holomorphic in z and t . We compute its derivative with respect to t . In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1, 0), d \rangle (1, 0).$$

The corresponding derivation of $\mathcal{O}(1)$ annihilates s , so that simply

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through $s \in W_D^0(-\frac{1}{2})$ there passes a horizontal local section of $\mathcal{W}(-\frac{1}{2})$. It is automatically a section of $\mathcal{W}^0(-\frac{1}{2})$. It exists in a neighborhood of D that is larger the slower the growth of s . Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection ∇ on $\mathcal{W}^0(-\frac{1}{2})$ defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\frac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of $\mathcal{W}^0(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{W}^0)$ on S has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{W}^0)$.

First proof. Divide S into two hemispheres S_1 and S_2 , and choose $\mathcal{O}(-\frac{1}{2})_i$ on S_i . On $S_1 \cap S_2$ one has locally an isomorphism $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$, defined up to sign, but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$: the monodromy is -1 . If there existed global horizontal projective sections of $\mathbb{P}(\mathcal{W}^0)$, there would exist horizontal sections f_1, f_2 of $\mathcal{W}^0(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would coincide up to sign. On $S_1 \cap S_2$

this would make it possible to normalize $\alpha : \mathcal{O}(-\frac{1}{2})_1 \rightarrow \mathcal{O}(-\frac{1}{2})_2$ by $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x -axis and D the y -axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D . By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{W}^0)$ passing through f , it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon|z|^2)). \quad (8.1)$$

If E' is another line, then by (4.1) the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon|z|^2)) \quad (8.2)$$

with suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$; hence in all directions except those for which az^2 is purely imaginary, g decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf, g decreases exponentially quadratically in all directions, hence is zero, and therefore $f = 0$.

9

In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. The commutator of d and $\bar{d}$ is

$$\langle d, \bar{d} \rangle = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \bar{d} \rangle > 1 \iff \operatorname{Im}(t) > 0.$$

The lines D such that, for d generating D , one has $\langle d, \bar{d} \rangle > 1$, therefore form a disk in S , called the disk attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ for which $\langle \cdot, \cdot \rangle$ is real on L form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, with $\operatorname{SL}(2, \mathbb{R})$ as the stabilizer of $\mathbb{R}^2$. Each defines a real structure $H_L = L \cdot U^1$ on H , hence a complex conjugation σ_L with fixed points LU^1 .

The disks in S likewise form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, the stabilizer of the one associated to $H_{\mathbb{R}}$ being $\operatorname{SL}(2, \mathbb{R})$. The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S .

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

10

The calculations of no. 3 admit the following interpretation. Let X be an open disk of S , corresponding to a real structure L , whence real forms H_L of H and $\operatorname{SL}(L)$ of $\operatorname{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ has an H_L -invariant pre-Hilbert structure; its completion $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$ is the irreducible unitary representation of H_L . One has

$$\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\frac{1}{2})^{\nabla=0}).$$

The double cover $\mathrm{SL}(L)^\sim$ of $\mathrm{SL}(L)$ acts on $\mathcal{O}(-\frac{1}{2})|_X$ compatibly with its action on $\mathcal{O}(-1)$; hence $\mathrm{SL}(L)^\sim$ acts on $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$. This is the action that normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)$ on H_L .

The subrepresentation $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ may be interpreted as the space of real analytic vectors for the action of $\mathrm{SL}(L)^\sim$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

11

Everything preceding generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle \cdot, \cdot \rangle$ real on $\mathbb{R}^{2n}$, replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle)),$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \bar{d} \rangle| > 1$ for $d \neq 0$ in D . The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ must be replaced by $\det(D)^{1/2}$.

12. Remark

Beside representations of H in which $\mathbb{C}^*$ acts by multiplication by λ , consider likewise representations in which the action is by λ^{-1} . One defines, as in no. 2, a holomorphic bundle $\mathcal{W}_-^0$ of such representations over S , and on $\mathcal{W}_-^0(-\frac{1}{2})$ one again has a connection, given by a heat equation.

On a connected simply connected open subset U of S , consider $W = \mathcal{W}^0(-\frac{1}{2})$, its connection, and the action of H . In $\Gamma(U, W)^{\nabla=0}$, for each $x \in U$ there is a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(e)$ for $f \in \mathcal{W}_x^0$. For each $x \notin U$, there exists a vector $v(x)$ fixed by D_x ; in the Schrödinger model, and for x corresponding to $E =$ the x -axis, it is the constant 1. It is unique up to a factor.

Fix the real structure $H_{\mathbb{R}}$, whence a real structure $S_{\mathbb{R}} \subset S$ (a circle), and a decomposition of $S - S_{\mathbb{R}}$ into two disks U_1, U_2 . Then:

- (a) The complex conjugate of $\Gamma(U_1, W)^{\nabla=0}$ is $\Gamma(U_2, W_-)^{\nabla=0}$.
- (b) The Hermitian form on $\Gamma(U_1, W)^{\nabla=0}$, invariant under $H_{\mathbb{R}}$, is reinterpreted as a bilinear pairing between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, invariant under H .

Question. Let C be a Jordan curve in S , separating $S - C$ into two open sets U_1 and U_2 . Does there exist a unique separating H -invariant bilinear form pairing $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, induced by a perfect duality between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$?

References

- [1] D. Shale, *Linear symmetries of free Boson fields*, Trans. Amer. Math. Soc. 103 (1962), 149–167.
- [2] J. von Neumann, *Die Eindeutigkeit der Schrödingerschen operatoren*, Ann. Mat. Pura Appl. (4) 104 (1931), 570–578.
- [3] A. Weil, *Sur certains groupes d'opérateurs unitaires*, Acta Math. 111 (1964), 143–211.

The Tame Symbol

Pierre Deligne

Working English translation. This is a faithful mathematical working translation preserving the article's numbered structure, formulas, and constructions. The accompanying French TeX layer is a cleaned working alignment layer.

1. Introduction

Let f and g be two meromorphic functions on a Riemann surface S . The *tame symbol* of f and g at $x \in S$ is

$$(f, g)_x = (-1)^{v_x(f)v_x(g)} \left(\frac{f^{v_x(g)}}{g^{v_x(f)}} \right) (x), \quad (1.1)$$

where $v_x(f)$ denotes the valuation of f at x , namely the order of the zero, or the negative of the order of the pole. The quotient appearing in (1.1) has valuation zero and hence can be evaluated at x . The definition is due to J. Tate; in the language of local symbols for the multiplicative group it gives, for compact S , the product formula

$$\prod_{x \in S} (f, g)_x = 1, \quad (1.2)$$

where almost all factors are equal to 1. This generalizes Weil reciprocity

$$f(\operatorname{div} g) = g(\operatorname{div} f)$$

for functions with disjoint divisors. If K is the fraction field of a discrete valuation ring with residue field k , formula (1.1) still defines a pairing

$$K^* \otimes K^* \longrightarrow k^*,$$

and, up to sign, this map can be interpreted in K -theory as the composite of the product $K_1(K) \otimes K_1(K) \rightarrow K_2(K)$ with the boundary $K_2(K) \rightarrow K_1(k)$.

Let D be the region in $\mathbf{C}$ bounded by a C^∞ curve Γ , and let f, g be meromorphic functions on a neighbourhood of D , without zeros or poles on Γ . The starting point of this article was a formula expressing

$$\prod_{x \in D} (f, g)_x$$

as the monodromy along Γ of a flat line bundle on Γ defined by the restrictions of f and g to Γ . Concretely, this is an integral formula for that product.

Following a suggestion of O. Gabber, the essential point is the construction, in the derived category of sheaves of abelian groups on a Riemann surface S , of a morphism of degree one

$$\mathcal{O}^* \otimes^L \mathcal{O}^* \longrightarrow \mathbf{C}^*[1]. \quad (1.3)$$

This morphism is given by an explicit formula at the level of complexes resolving $\mathcal{O}^*$ and $\mathbf{C}^*$. It therefore has a local interpretation: to two invertible holomorphic functions f and g it attaches a $\mathbf{C}^*$ -torsor, equivalently a flat line bundle, denoted (f, g) .

In arbitrary dimension there is an analogous morphism, with values in the complex

$$\mathcal{O}^* \xrightarrow{d\log} \Omega^1,$$

which is quasi-isomorphic to $\mathbf{C}^*$ in dimension one but not in higher dimension. It attaches to two invertible holomorphic functions f, g a line bundle (f, g) with connection, not necessarily integrable. In fact this morphism is the one defining the product in Beilinson-Deligne cohomology, $\mathbf{Z}(1) \otimes \mathbf{Z}(1) \rightarrow \mathbf{Z}(2)$; the corresponding interpretation in terms of Hodge theory is described in Section 3.

The same constructions exist on C^∞ manifolds, where $\mathcal{O}^*$ is replaced by the sheaf of invertible C^∞ functions and $\mathcal{O}^* \rightarrow \Omega^1$ by its differentiable analogue. Less regularity is also possible.

For a commutative complex analytic group G , (1.3) admits the generalization

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1], \quad (1.4)$$

which attaches a flat G -torsor $(f, g]$ to an invertible holomorphic function f and a holomorphic map g with values in G on a Riemann surface. Its monodromy is analogous to the local symbols of class field theory. In particular, if $\mathcal{L}$ is an invertible sheaf trivialized at infinity and g is a holomorphic map to G , one obtains a symbol $(\mathcal{L}, g] \in G$ with $(\mathcal{O}(x), g] = g(x)$.

For the circle S^1 , the map

$$(f, g) \longmapsto \text{monodromy}(f, g)$$

identifies, in a sense made precise below, the multiplicative group of invertible functions on S^1 with its own Cartier dual. This autoduality is skew-symmetric. It is an analytic analogue of the Contou-Carrere result that the functor sending a commutative ring A to $(A[[T]][T^{-1}])^*$ is its own Cartier dual. A twisted version with coefficients in a local system of free finite-rank $\mathbf{Z}$ -modules on S^1 is also proved.

If S is a Riemann surface with boundary a circle, the boundary values of invertible holomorphic functions form a maximal isotropic subspace of the group of invertible smooth functions on the boundary. Variants are proved for several boundary circles and for local systems. These results are parallel to the autoduality of the Jacobian, and of Prym varieties.

Throughout, Tate twists such as $\mathbf{Z}(1) = 2\pi i \mathbf{Z}$ are needed to make the statements natural and independent of the choice of a square root i of -1 in $\mathbf{C}$.

The last section applies these tools to a preliminary construction needed for holomorphic conformal field theories attached to certain orbifolds of the form “torus modulo a finite group”. Let B be an even positive definite bilinear form on $\mathbf{Z}^n$, and let S be an oriented circle. From B and an auxiliary datum one obtains a central extension of the group of smooth maps $S \rightarrow \mathbf{R}^n/\mathbf{Z}^n$ by $\mathbf{C}^*$, and a compatibility with gluing along surfaces with boundary.

1 The main construction

Let X be a complex analytic variety. We write $\mathcal{O}$ for the sheaf of holomorphic functions, $\mathcal{O}^*$ for the invertible functions, and $G(\mathcal{O})$ for the sheaf of holomorphic maps to a commutative complex Lie group G . The exponential sequence for G is written in the form

$$0 \longrightarrow G(\mathbf{C}) \longrightarrow G(\mathcal{O}) \xrightarrow{d\log} \Omega^1 \otimes \text{Lie}(G) \longrightarrow 0 \quad (2.1.1)$$

when X is one-dimensional and G is locally represented in logarithmic coordinates.

For a complex $[A \rightarrow B]$ placed in degrees $n, n+1$ we write $[A \rightarrow B]_n$. The fundamental morphism is obtained from the elementary formula

$$[\mathcal{O}^* \rightarrow \Omega^1]_0 \otimes^L G(\mathcal{O}) \longrightarrow G[1]. \quad (2.2.1)$$

At the level of local logarithms, it is given by the expression

$$(f, g) \longmapsto \exp \left(\frac{1}{2\pi i} \int \log(f) d \log(g) \right), \quad (2.2.2)$$

with the usual correction terms on overlaps. Passing to cohomology, (2.2.1) gives a product

$$H^i(X, \mathcal{O}^*) \times H^j(X, G(\mathcal{O})) \longrightarrow H^{i+j+1}(X, G). \quad (2.3.1)$$

For $i = j = 0$ this attaches to local functions f, g the torsor $(f, g]$.

The construction may be written using Čech cocycles. Let $\{U_i\}$ be an open cover, choose branches $\log f_i$ of f , and represent g locally. The change of branch $\log f_i - \log f_j \in 2\pi i \mathbf{Z}$ supplies the transition functions of a G -torsor by applying these integral changes to the periods of g . This Čech description is often more explicit than the derived-category one, but it is the same construction.

On a C^∞ manifold the construction is identical after replacing holomorphic functions by smooth invertible functions and holomorphic differentials by smooth one-forms. For a Riemann surface S , since $[\mathcal{O}^* \rightarrow \Omega^1]$ is quasi-isomorphic to $\mathbf{C}^*$, the construction becomes a pairing

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1] \quad (2.6.1)$$

which yields flat torsors.

Let $x \in S$, let f be invertible on $S - \{x\}$ with prescribed divisor at x , and let $g : S - \{x\} \rightarrow G$. The monodromy of $(f, g]$ around x is the local symbol. For $G = \mathbf{G}_m$ it is the tame symbol (1.1). For $G = \mathbf{G}_a$ one obtains the residue pairing

$$(f, g]_x = \text{res}_x(g d \log f). \quad (2.8.1)$$

For $G = \mathbf{G}_m$ we write (f, g) instead of $(f, g]$. The symbol is bimultiplicative and satisfies the product formula on compact Riemann surfaces.

Let $\mathcal{L}$ be an invertible sheaf on S trivialized at infinity. Applying the construction to a divisor representing $\mathcal{L}$ gives a global symbol

$$(\mathcal{L}, g] \in G, \quad (2.10.1)$$

characterized by $(\mathcal{O}(x), g] = g(x)$ and by multiplicativity in divisors. The case of cohomology with compact support gives a parallel construction when $i = j = 1$.

There is also a relative version. For a family of Riemann surfaces and holomorphic functions varying over a base, the torsors $(f, g]$ vary functorially and are compatible with base change. The local monodromy formula survives in families.

2 The case of the multiplicative group

Take $G = \mathbf{G}_m$. The product (2.2.1) becomes the product in Deligne-Beilinson cohomology

$$\mathbf{Z}(1)_{\mathcal{D}} \otimes^L \mathbf{Z}(1)_{\mathcal{D}} \longrightarrow \mathbf{Z}(2)_{\mathcal{D}}, \quad (3.1.1)$$

where $\mathbf{Z}(1) = 2\pi i\mathbf{Z}$ and $\mathbf{Z}(2) = (2\pi i)^2\mathbf{Z}$. Explicitly, the line bundle with connection (f, g) is represented locally by logarithms of f and g and by the connection one-form

$$\nabla = d + \frac{1}{2\pi i} \log(f) d \log(g). \quad (3.1.2)$$

When X is a Riemann surface the connection is flat and its monodromy is the tame symbol.

The Čech formula reads as follows. Choose logarithms $u_i = \log f$ and $v_i = \log g$ on U_i . On intersections the differences $u_i - u_j$ and $v_i - v_j$ lie in $2\pi i\mathbf{Z}$. The transition functions of (f, g) and the local connection forms are assembled from

$$\frac{1}{2\pi i} u_i dv_i, \quad \exp\left(\frac{1}{2\pi i} (u_i - u_j) v_j\right), \quad (3.2.1)$$

with the standard cocycle correction on triple intersections. The same formula with parameters gives the relative torsor.

The construction is bimultiplicative. There are canonical isomorphisms

$$(f_1 f_2, g) \simeq (f_1, g) \otimes (f_2, g), \quad (3.7.1)$$

$$(f, g_1 g_2) \simeq (f, g_1) \otimes (f, g_2). \quad (1)$$

It is skew-symmetric in the sense that there is a canonical isomorphism

$$(f, g) \otimes (g, f) \simeq \mathbf{1}. \quad (3.8.1)$$

For $f = -1$ the canonical trivialization in (3.8.1) must be treated with the sign convention encoded in the Tate twist; otherwise one misses the contribution of the chosen square root of -1 .

By bilinearity, one may extend (f, g) to sections of a local system. Let A be a local system of free finite-rank $\mathbf{Z}$ -modules, $B : A \otimes A \rightarrow \mathbf{Z}$ a bilinear form, and let f be a section of $\mathcal{O}^* \otimes A$. The expression

$$(f, f)_B \quad (3.10.1)$$

is defined by combining the pairings of the components with coefficients given by B . If B is symmetric and even, the construction is independent of ordering. If B is symmetric but not necessarily even, the parity term

$$B(x, x) \pmod{2} \quad (3.11.1)$$

controls the additional sign.

Let S be an oriented circle and A a local system of free finite-rank $\mathbf{Z}$ -modules on S . Choosing $i \in \mathbf{C}$ identifies $\mathbf{Z}$ with $\mathbf{Z}(1)$ by $1 \mapsto 2\pi i$. The exponential sequence gives exact sequences of sheaves on S :

$$0 \longrightarrow A \longrightarrow \mathcal{C}^\infty \otimes A \xrightarrow{\exp(2\pi i \cdot)} (\mathcal{C}^\infty)^* \otimes A \longrightarrow 0, \quad (3.13.1)$$

$$0 \longrightarrow A \longrightarrow \mathcal{C} \otimes A \longrightarrow \mathbf{C}^* \otimes A \longrightarrow 0. \quad (2)$$

These sequences will be used in the duality theorem of Section 4.

3 Duality

This section gives analytic analogues of the autoduality of the Jacobian and of the Contou-Carrere symbol. For an oriented C^∞ circle S the exponential sequence

$$0 \longrightarrow 2\pi i \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0 \quad (4.1.1)$$

and the complex

$$0 \longrightarrow \mathbf{C}^* \longrightarrow \mathcal{O}^* \xrightarrow{d\log} \Omega^1 \longrightarrow 0 \quad (4.1.2)$$

identify the group of smooth invertible functions, up to the correct completion, with its Cartier dual.

More generally, for a local system A of free $\mathbf{Z}$ -modules, the pairing of Section 3 gives a perfect pairing between the group of invertible smooth functions with values in A and the corresponding dual group with values in $A^\vee(1)$. The duality is skew-symmetric and functorial in A .

One treats the relevant cohomology groups as topological abelian groups. For a locally convex topological vector space the dual means the continuous character group. Exactness is understood in the topological sense. The pairings are computed by integrating logarithms and logarithmic differentials along the circle; in a local coordinate x one encounters expressions of the form

$$\int_S a(x) d\log f(x) \quad (4.6.1)$$

and, for the skew form, expressions of the form

$$\int_S \log f d\log g. \quad (4.6.2)$$

If A is a local system on S , the exact sequence

$$0 \rightarrow H^0(S, A) \rightarrow H^0(S, \mathcal{O} \otimes A) \rightarrow H^0(S, \Omega^1 \otimes A) \rightarrow H^1(S, A) \rightarrow 0 \quad (4.7.1)$$

serves to write the pairing explicitly and to show its non-degeneracy. In the basic example $S = U(1)$ and $A = \mathbf{Z}$, the group of invertible smooth functions decomposes into winding number and exponentials of smooth functions; the duality exchanges these two parts as expected.

Let S now be a compact Riemann surface with boundary. Boundary values of holomorphic invertible functions form an isotropic subgroup for the boundary pairing. The image is maximal isotropic. This is the analytic counterpart of the assertion that the Jacobian of a compact Riemann surface is principally polarized.

With coefficients in a local system A , and a positive definite bilinear form B on A , the boundary pairing is written

$$\langle f, g \rangle_B = (f, g)_B. \quad (4.16.1)$$

The proof shows that $\log(f, f)_B$ is the L^2 -norm of a harmonic representative. Hence for positive definite B the corresponding isotropic condition is also a positivity statement. For the annulus one obtains a Fourier-expansion model: if the two boundary components are related by the modulus of the annulus, the pairing is expressed as a convergent sum over Fourier modes.

4 Construction of central extensions

Let S be a compact Riemann surface with boundary and A a local system on S endowed with an even positive definite symmetric bilinear form B . The objective is to construct central extensions of groups of boundary values by $\mathbf{C}^*$, using the torsors and gerbes associated with the symbol.

We use the language of torsors and gerbes. A G -torsor is a sheaf locally isomorphic to G with its simply transitive G -action. A gerbe banded by G is a stack locally non-empty and locally connected, with automorphism group G ; it represents classes in $H^2(-, G)$. The torsor associated with (f, g) is thus the H^1 incarnation of the symbol, and the gerbe construction is its H^2 analogue.

A central extension of a sheaf of groups L by G is a G -torsor over L equipped with a multiplication lifting that of L , with G central. Equivalently, it is given locally by a two-cocycle, with the usual coboundary equivalence. In categorical language, the same data are encoded by a gerbe with a compatible group law.

Following an idea of Grothendieck, the auxiliary data needed to construct the extension are kept in a sheaf $\mathcal{H}^*$. A section consists of a lattice with bilinear form and of the choices required to make the signs and Tate twists coherent. If $k = (\mathbf{C}, s)$ is such an auxiliary datum, one defines an extension

$$1 \longrightarrow \mathbf{C}^* \longrightarrow \tilde{L}_k \longrightarrow L \longrightarrow 1. \quad (5.12.1)$$

The construction is local on the boundary and compatible with restriction to boundary components.

If $k' = (A', B')$ and $k'' = (A'', B'')$ are two data, the extension associated with $k' + k''$ is obtained from those associated with k' and k'' by the contracted product. The commutator in the extension is the symbol obtained from the boundary pairing:

$$[\tilde{f}, \tilde{g}] = (f, g)_B. \quad (5.16.1)$$

For a circle lying on the boundary of a surface, the data of Section 5.1 therefore provide a central extension of the boundary loop group. These extensions are the analytic input for the construction of holomorphic conformal field theories attached to torus orbifolds.

References

- [1] A. Beilinson, Higher regulators and values of L -functions.
- [2] J. Giraud, Cohomologie non abelienne.
- [3] J. Tate, Residues of differentials on curves.

Real Mixed Hodge Structures

Pierre Deligne

English working translation; formulas, filtrations, and numbering retained.

The real mixed Hodge structures of §2 form a rigid abelian tensor category, and the functor “underlying real vector space” is a fiber functor. The functor Gr^W is another one. We compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposite filtrations

1.1

This paragraph is a variation on §1.2 of P. Deligne, *Théorie de Hodge II*, Publ. Math. IHES 40, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \bar{F}$. Here “filtration” means finite filtration; W is increasing, and F and $\bar{F}$ are decreasing. The condition of opposition is

$$\mathrm{Gr}_n^W \mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q A = 0 \quad \text{for } n \neq p + q.$$

For a bigraded object $B = \bigoplus B^{p,q}$, we write $W, F_W, \bar{F}_W$ for the three opposite filtrations

$$W_n B = \bigoplus_{p+q \leq n} B^{p,q}, \quad F_W^i B = \bigoplus_{p \geq i} B^{p,q}, \quad \bar{F}_W^i B = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, saying that $W, F, \bar{F}$ are opposite on A is equivalent to saying that, for each n , $\mathrm{Gr}_n^W A$ has a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations induced by F and $\bar{F}$ on $\mathrm{Gr}_n^W A$ are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, p \geq i} A_W^{p,q}, \quad \bar{F}^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, q \geq i} A_W^{p,q}.$$

Put

$$A_F^{p,q} := W_{p+q} A \cap F^p A \cap \left(\bar{F}^q A + \sum_{i>0} W_{p+q-i} A \cap \bar{F}^{q-i+1} A \right),$$

and, interchanging the roles of $F, \bar{F}$ and of p, q ,

$$A_{\bar{F}}^{p,q} := W_{p+q} A \cap \bar{F}^q A \cap \left(F^p A + \sum_{i>0} W_{p+q-i} A \cap F^{p-i+1} A \right).$$

For $p + q = n$, the projection from $W_n A$ to $\mathrm{Gr}_n^W A$ induces isomorphisms

$$A_F^{p,q} \simeq A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \simeq A_W^{p,q}.$$

It follows that the $A_F^{p,q}$, respectively the $A_{\bar{F}}^{p,q}$, form a bigrading of A . The sums of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

are bifiltered isomorphisms

$$a_F : (A; W, F) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, F_W), \quad a_{\bar{F}} : (A; W, \bar{F}) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, \bar{F}_W).$$

Lemma 1.1. *The automorphism $d = a_F a_{\overline{F}}^{-1}$ of $\mathrm{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r < p \\ s < q}} A_W^{r,s}. \quad (1.1.1)$$

Proof. Passing to the associated graded, the isomorphisms a_F and $a_{\overline{F}}$ induce the identity on $\mathrm{Gr}^W A$. Thus $\mathrm{Gr}^W(d) = 1$. For a subset $I \subset \mathbb{Z}^2$ and a bigraded object B , write $B_I = \bigoplus_{(r,s) \in I} B^{r,s}$. Let

$$I(p, q) = \{(r, s) : r + s \leq p + q \text{ and } ((r, s) = (p, q) \text{ or } r < p)\},$$

$$J(p, q) = \{(r, s) : r + s \leq p + q \text{ and } ((r, s) = (p, q) \text{ or } s < q)\},$$

$$K(p, q) = \{(r, s) : r + s \leq p + q \text{ and } r < p \text{ and } s < q\}.$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects, the first subobject is $A_{FI(p,q)}$ and the second is $A_{\overline{F}J(p,q)}$. Hence $dA_W^{p,q}$ lies in the intersection corresponding to $I(p, q) \cap J(p, q) = K(p, q) \cup \{(p, q)\}$. Since $\mathrm{Gr}^W(d) = 1$, this gives exactly (1.1.1). $\square$

Proposition 1.1. *The functor $A \mapsto \mathrm{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \overline{F}$ to the category of objects of $\mathcal{A}$ endowed with a finite bigrading and with an automorphism d satisfying (1.1.1).*

The isomorphism $a_F : A \rightarrow \mathrm{Gr}^W A$ respects W , sends F to F_W , and sends $\overline{F}$ to $d^{-1}\overline{F}_W$. Thus the functor Gr^W has the inverse

$$s : (B = \bigoplus B^{p,q}, d) \mapsto (B, W, F_W, d^{-1}\overline{F}_W),$$

with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \mathrm{Gr}^W A$. Conversely, starting from a bigraded object B with an automorphism d , define $W, F, \overline{F}$ as above. If d satisfies (1.1.1), the evident map $B \rightarrow \mathrm{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \mathrm{Gr}^W \circ s$.

Remark 1.1. *If 2 is invertible in the ground field, then d has a unique square root $d^{1/2}$ still satisfying (1.1.1). Put*

$$a := d^{1/2} a_F = d^{-1/2} a_{\overline{F}}.$$

Then $a_{\overline{F}} = d^{1/2} a$ and $a_F = d^{-1/2} a$; thus a sends F , respectively $\overline{F}$, to

$$d^{1/2}(F_W), \quad d^{-1/2}(\overline{F}_W)$$

on $\mathrm{Gr}^W A$. The isomorphism a is the same for $(A; W, F, \overline{F})$ and for $(A; W, \overline{F}, F)$.

1.4

The constructions are compatible with tensor products. If

$$\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{A}$$

is a biadditive exact functor, it is extended to filtered objects by

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a A_1 \otimes F^b A_2 \subset A_1 \otimes A_2.$$

The associated-graded functor is compatible with $\otimes$, and, for bifiltered objects, the canonical isomorphism

$$\mathrm{Gr}^W \mathrm{Gr}^F(A_1 \otimes A_2) \simeq \mathrm{Gr}^W \mathrm{Gr}^F(A_1) \otimes \mathrm{Gr}^W \mathrm{Gr}^F(A_2)$$

sends the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2 . For vector spaces this follows from the fact that every bifiltration underlies a bigrading. Hence, if $(W_i, F_i, \overline{F}_i)$ are three opposite filtrations on A_i , then $(W_1 \otimes W_2, F_1 \otimes F_2, \overline{F}_1 \otimes \overline{F}_2)$ are again opposite on $A_1 \otimes A_2$. The bigradings, the isomorphisms $a_F, a_{\overline{F}}$, and the automorphism d are all compatible with $\otimes$.

1.5

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k , and let $\mathcal{C}$ be the category of vector spaces endowed with three opposite filtrations $W, F, \overline{F}$. This is a rigid abelian tensor category with $\text{End}(1) = k$. It has the fiber functor ω_0 , the underlying vector space, hence it is neutral Tannakian. It also has the fiber functor $\text{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\overline{F}}$ are isomorphisms of fiber functors $\omega_0 \rightarrow \omega_W$. Put

$$\mathcal{G} = \text{Aut}(\omega_W).$$

The functorial automorphism d defines a k -point of the pro-algebraic group $\mathcal{G}$. The functorial bigrading of $\text{Gr}^W(V)$ by the $V^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow \mathcal{G}.$$

It is a section of the projection

$$\text{pr} : \mathcal{G} \longrightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion of the category of bigraded vector spaces in $\mathcal{C}$. We normalize α by

$$\alpha(\lambda, \mu)v = \lambda^p \mu^q v \quad (v \in V^{p,q}).$$

By Proposition 1.1, $\mathcal{G}$ is also the algebraic group of automorphisms of the fiber functor on the category of bigraded vector spaces endowed with an automorphism d satisfying (1.1.1). We now compute it when k has characteristic zero.

Let $\mathfrak{l}$ be the Lie algebra over k freely generated by elements

$$D^{i,j} \quad (i, j < 0),$$

with the unique bigrading in which $D^{i,j}$ has bidegree (i, j) . The $W_n \mathfrak{l}$ are ideals, and the quotients $\mathfrak{l}/W_n \mathfrak{l}$ are nilpotent Lie algebras. Let $\mathcal{U}$ be the pro-algebraic group which is the projective limit of the unipotent groups

$$\exp(\mathfrak{l}/W_n \mathfrak{l}).$$

Let $\mathbb{G}_m \times \mathbb{G}_m$ act on $\mathfrak{l}$ so that (λ, μ) acts on bidegree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects W and therefore gives an action of $\mathbb{G}_m \times \mathbb{G}_m$ on $\mathcal{U}$.

Construction 1.1. *The group $\mathcal{G}$ is the semidirect product*

$$\mathcal{G} = (\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U}.$$

Indeed, giving an action of this semidirect product on a finite-dimensional vector space V is equivalent to giving an action of $\mathbb{G}_m \times \mathbb{G}_m$, hence a bigrading, together with a compatible nilpotent action of $\mathfrak{l}$. Equivalently, it is the datum of endomorphisms $D^{i,j}$ of bidegree (i, j) , or of their sum

$$D = \sum D^{i,j},$$

with

$$D(V^{p,q}) \subset \bigoplus_{r < p, s < q} V^{r,s}.$$

Writing $d = \exp(D)$, this is precisely the datum of a bigrading and of an automorphism satisfying (1.1.1). The construction is compatible with tensor products, giving the required isomorphism

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U} \xrightarrow{\sim} \mathcal{G} = \text{Aut}(\omega_W).$$

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space endowed with an increasing filtration W , whose complexification $V_{\mathbb{C}}$ is endowed with a decreasing filtration F , such that the filtration $W_{\mathbb{C}}$, the filtration F , and its complex conjugate $\bar{F}$ form a system of three opposite filtrations.

Via the dictionary between real vector spaces and complex vector spaces with a real structure, giving a real mixed Hodge structure is equivalently giving a complex vector space with three opposite filtrations $W, F, \bar{F}$ and with a complex conjugation respecting W and interchanging F and $\bar{F}$. By Proposition 1.1, the functor Gr^W identifies this category with the category of bigraded complex vector spaces

$$V = \bigoplus V^{p,q}$$

equipped with an automorphism d satisfying (1.1.1) and a complex conjugation σ interchanging $V^{p,q}$ and $V^{q,p}$, for which the complex conjugate of d is d^{-1} :

$$\sigma d \sigma^{-1} = d^{-1}.$$

Let $\mathcal{H}$ be the category of real mixed Hodge structures. It is a rigid abelian tensor category with $\text{End}(1) = \mathbb{R}$. It has the fiber functor ω_0 , the underlying real vector space, hence is neutral Tannakian. It also has the fiber functor $\text{Gr}^W =: \omega_W$. With the notation of Remark 1.1, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \text{Gr}^W(V_{\mathbb{C}})$$

respects the real structures on both sides, since it depends symmetrically on F and $\bar{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, put $k = \mathbb{C}$, and let $\mathcal{G}_{\mathbb{R}}$ be the real form of $\mathcal{G}$ defined by the following complex conjugation:

- (a) on $\mathbb{G}_m \times \mathbb{G}_m$, it is $(\lambda, \mu) \mapsto (\bar{\mu}, \bar{\lambda})$;
- (b) on $\mathcal{U}$, it is the one compatible with the complex conjugation of $\mathfrak{l}$ given by

$$D^{i,j} \longmapsto -D^{j,i}.$$

As in §1.1, giving an action of $\mathcal{G}_{\mathbb{R}}$ on a real vector space V is equivalent to giving:

- (a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another;
- (b) an automorphism d of $V_{\mathbb{C}}$ such that $\bar{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{i < p, j < q} V^{i,j}.$$

Proposition 2.1. *The fiber functor ω_W induces an equivalence between $\mathcal{H}$ and the category of representations of $\mathcal{G}_{\mathbb{R}}$. Thus*

$$\mathcal{G}_{\mathbb{R}} = \text{Aut}(\omega_W).$$

The inverse functor is

$$(V, \text{ action of } \mathcal{G}_{\mathbb{R}}) \longmapsto (V, \text{ bigrading of } V_{\mathbb{C}}, d) \longmapsto (V, W, F),$$

where W is the real filtration whose complexification is the weight filtration attached to the bigrading of $V_{\mathbb{C}}$, and

$$F = d^{-1/2}(F_W)$$

on the complexification. The isomorphism a gives the functorial identification

$$\phi \circ \text{Gr}^W \xrightarrow{\sim} \text{Id}.$$